Chapter 6

State feedback

In the previous chapter we faced the problem of determining a control law that brings the state of the system from a given initial point to a given final point. The solution that we computed is "open loop", in the sense that the control law $\mathbf{u}(\cdot)$ has been determined only based on these two given values (initial and final point) plus the characteristics of the system. In other words, the control law computed before does not account for the dynamics of the system while the input is applied to the system. Every time we adopt an open loop control action, the generation of the input signal is delegated to a device that is external with respect to the system, in the sense that, while operating, this device does not have knowledge of how the system is evolving.

To determine control laws accounting for the system evolution while the law is being applied requires using information about the current values of the state or the output of the system, and hence to consider closed loop solutions, obtained by adopting a state- or output-feedback. The input thus obtained can be computed either using this knowledge in a static way, or by elaborating the control law through an external dynamic system. In the first case the feedback is called "static feedback", while in the second it is called "dynamic feedback" (from the state or from the output, respectively).

Referring to the case of linear systems, the most common control problems can be classified as follows:

(i) *regulation problems*, where the system initial condition is different from zero because of the effects of some disturbances, and the aim of the control law is to bring the system back to its rest position with a predefined convergence speed;

(ii) *tracking problems*, where the output of the system is instead required to approximate a certain trajectory, according to some approximation criteria.

The methodology we will introduce in this chapter can then be classified as a static state-feedback to address regulation problems.

6.1 State equations for feedback systems

Consider a discrete- or continuous-time system $\Sigma = (F, G, H)$ and assume that all the state entries are available for the purpose of designing a feedback law. The simplest feedback law is the one where the inputs $u_i(t),\ i = 1, 2, \ldots, m$, are computed as linear combinations of the components of the state of the system at the same time t plus some other "external" inputs $v_i(t)$, i.e.,

$$u_i(t) = k_{i1}x_1(t) + k_{i2}x_2(t) + \ldots + k_{in}x_n(t) + v_i(t), \quad i = 1, 2, \ldots, m. \tag{6.1}$$

In other words, the simplest feedback law can be written in matrix form as

$$\mathbf{u}(t) = K\mathbf{x}(t) + \mathbf{v}(t), \tag{6.2}$$

where the *feedback matrix* $K = [k_{ij}]$ has sise $m \times n$. To fix the ideas, consider the continuous-time system

$$\begin{aligned} \dot{\mathbf{x}}(t) &= F\mathbf{x}(t) + G\mathbf{u}(t) \\ \mathbf{y}(t) &= H\mathbf{x}(t). \end{aligned}$$

Substituting (6.2) into $\mathbf{u}(t)$ we obtain

$$\begin{aligned} \dot{\mathbf{x}}(t) &= (F + GK)\mathbf{x}(t) + G\mathbf{v}(t) \\ \mathbf{y}(t) &= H\mathbf{x}(t), \end{aligned} \tag{6.3}$$

i.e., the new state-space model $\Sigma^{(K)} = (F + GK, G, H)$.

A special case of state feedback is the one that can be obtained considering a static feedback from the output, i.e., a feedback law of the type

$$\mathbf{u}(t) = \bar{K}\mathbf{y}(t) + \mathbf{v}(t), \tag{6.4}$$

where $\bar{K} \in \mathbb{R}^{m \times p}$. Substituting $\mathbf{y}(t)$ into (6.4) we indeed have

$$\mathbf{u}(t) = \bar{K}H\mathbf{x}(t) + \mathbf{v}(t), \tag{6.5}$$

which is a special case of (6.2) with $K = \bar{K}H$.
Notice that, as $\bar{K}$ varies, the matrix $\bar{K}H$ describes the subspace of the $m \times n$ matrices whose rows are linear combinations of the rows of H. In particular, for single-input single-output systems (for which $m = p = 1$) the matrix $\bar{K}$ in (6.4) is a scalar, thus the matrix $\bar{K}H$ in (6.5) is a row vector that is proportional to H, while in (6.2) the matrix K is an arbitrary row vector with n components. As one may realise (and as we will see this in the following example), having n free parameters instead of only one makes state feedback significantly more effective in modifying the dynamics of Σ with respect to static output feedback.

Example 6.1.1 Consider the problem of controlling the angular position of a mechanical shaft, connected to the axis of a DC motor. Our goal is to design a feedback controller such that if the shaft is moved from its nominal position θ_d then it goes back to that position as fast as possible. A first control scheme consists in applying an input to the motor so that its exerted torque $u(t)$ at time t is proportional to the position error $e(t)$ defined as the difference between the nominal position θ_d and the actual angular position $\theta(t)$ of the shaft.

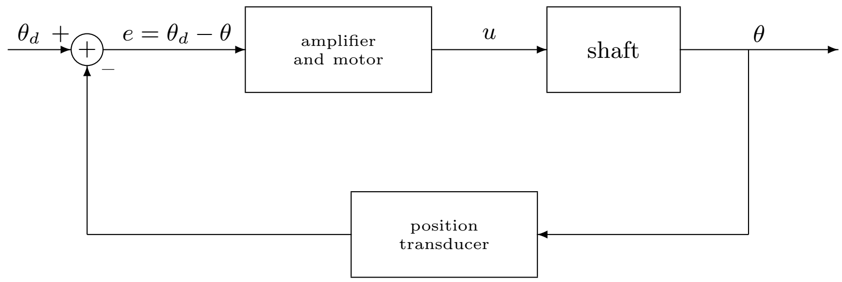


Figure 6.1.1

Suppose then that the time constants of the transducer, measuring the actual angular position $\theta(t)$, and of the motor are negligible with respect to the time constant of the load applied to the motor. Denoting by J the rotational inertia of the shaft, by β its viscous friction coefficient, and by k the proportionality constant between the position error e and the torque u exerted by the motor, the dynamics of the shaft can be written as

$$J\ddot{\theta} = -\beta\dot{\theta} + u \tag{6.6}$$
$$u = ke. \tag{6.7}$$

Setting then

$$\begin{aligned} x_1 &= -e = \theta - \theta_d \\ x_2 &= \dot{x}_1 = \dot{\theta} \end{aligned}$$

and assuming that the output of the overall system is the angular position θ, the model becomes

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{\beta}{J}x_2 + \frac{1}{J}u \\ y &= x_1 + \theta_d. \end{aligned}$$

Substituting the error e in (6.7) with $-x_1$ we get

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 \\ -\frac{k}{J} & -\frac{\beta}{J} \end{bmatrix}\mathbf{x} \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix}\mathbf{x} + \theta_d. \end{aligned} \tag{6.8}$$

The characteristic polynomial of the global system is thus $s^2 + s\frac{\beta}{J} + \frac{k}{J}$, and varying k in $\mathbb{R}_+$ we get the (positive) root locus shown in Figure 6.1.2.

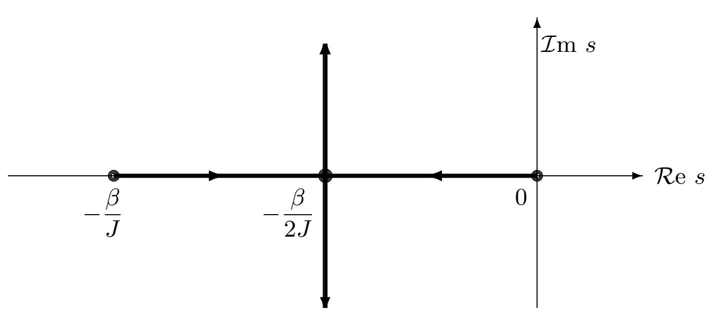


Figure 6.1.2

As it may be seen from the figure, at least one of the eigenvalues of the closed-loop system has always real part that is not smaller than $-\beta/2J$. This, in practice, constrains the speed of the overall system.

A more complex scheme consists in measuring, through a dedicated transducer, also the angular velocity of the shaft $\dot{\theta}$. With this additional information it is indeed possible to let the torque u applied by the motor be a linear combination of both the error in the angular position and in the angular speed of the shaft. In other words, it is now possible to set

$$u \;=\; k_1 e + k_2\dot{\theta} \;=\; -k_1x_1 + k_2x_2 \;=\; K\mathbf{x} \qquad \text{with } K = [\,-k_1 \quad k_2\,].$$

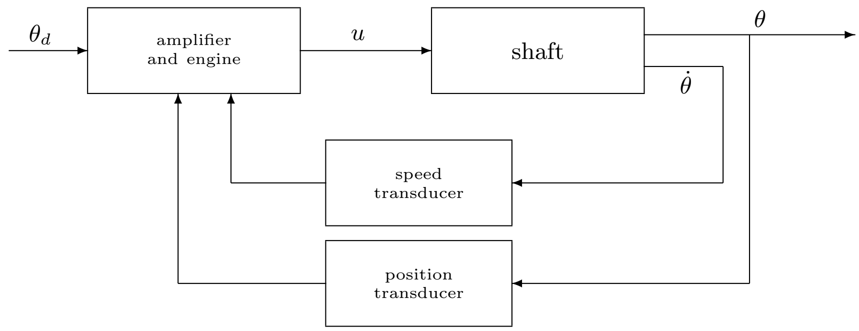


Figure 6.1.3

In this case the state update equations become

$$\dot{\mathbf{x}} \;=\; \begin{bmatrix} 0 & 1 \\ -\dfrac{k_1}{J} & -\dfrac{\beta - k_2}{J} \end{bmatrix}$$

$$y \;=\; [\;1 \quad 0\;]\,\mathbf{x} + \theta_d \tag{6.9}$$

so that the characteristic polynomial of the overall closed loop system is now $s^2 + \dfrac{\beta - k_2}{J}s + \dfrac{k_1}{J}$. Varying k_1 and k_2 in $\mathbb{R}$ we see that the zeros of the polynomial can be arbitrary. Importantly, one should notice how introducing an additional sensor allows more flexibility in the control design problem.

Exercise 6.1.1 Given the matrix $H \in \mathbb{R}^{p\times n}$, prove that

- if H has rank ν, then the space of the matrices $\bar{K}H$ in (6.5), as $\bar{K}$ varies, has dimension $m\nu$;
- a necessary and sufficient condition for a generic matrix $K \in \mathbb{R}^{m\times n}$ to factorise as $K = \bar{K}H$, for a suitable choice of $\bar{K}$ is that H has rank n.

Adopting a feedback law like (6.2) modifies the structure of the system, by changing the pair (F, G) into the pair $(F + GK, G)$. Our aim is now twofold: to determine which properties of the system Σ are preserved by any state feedback law as K varies, and to understand to which extent the properties of Σ can be modified by K.

As for the properties that are preserved by a state feedback law like (6.2), we prove the following two claims:

Proposition 6.1.1 [Invariance of the reachable subspace] *The reachable subspaces of $\Sigma = (F, G, H)$ and of $\Sigma^{(K)} = (F + GK, G, H)$ coincide for every $K \in \mathbb{R}^{m\times n}$. In particular, $\Sigma^{(K)}$ is reachable if and only if Σ is reachable.*

Proof Independently of K, let X^R and $X^{R,(K)}$ be the reachable subspaces of Σ and of $\Sigma^{(K)}$. These subspaces are, respectively, the smallest F-invariant and the smallest $(F + GK)$-invariant subspaces that contain the image of G. Therefore

$$(F + GK)X^R \subseteq FX^R + \mathrm{Im}G \subseteq X^R + \mathrm{Im}G = X^R. \tag{6.10}$$

Consequently, X^R is $(F+GK)$-invariant and it includes the image of G. Thus it also includes $X^{R,(K)}$, which is the smallest subspace satisfying these two properties. This implies that $X^R \supseteq X^{R,(K)}$, i.e., that the reachable subspace of the closed loop system is always included in the reachable subspace of the open loop system.

At the same time $\Sigma = (F, G, K) = (F + GK + G(-K), G, H)$ can be obtained from $\Sigma^{(K)} = (F + GK, G, H)$ by using the state feedback matrix $-K$. Thus the space $X^{R,(K)}$ includes X^R and therefore coincides with it. ■

Proposition 6.1.2 [Invariance of the spectrum of the non-reachable subsystem] *Let $\Sigma = (F, G, H)$ be in its standard reachability form, i.e.,*

$$\begin{aligned} F &= \begin{bmatrix} F_{11} & F_{12} \\ 0 & F_{22} \end{bmatrix}, \quad G = \begin{bmatrix} G_1 \\ 0 \end{bmatrix}, \quad \text{with } (F_{11}, G_1) \text{ reachable} \\ H &= [\, H_1 \quad H_2 \,] \end{aligned} \tag{6.11}$$

Then $\Sigma^{(K)} = (F+GK, G, H)$ is also in standard reachability form, and it shares the same sub-matrix F_{22} as Σ.

Proof Partition the feedback matrix K according to the partition of F, i.e.,

$$K = [\, K_1 \quad K_2 \,].$$

Then

$$F + GK = \begin{bmatrix} F_{11} & F_{12} \\ 0 & F_{22} \end{bmatrix} + \begin{bmatrix} G_1 \\ 0 \end{bmatrix} [\, K_1 \quad K_2 \,] = \begin{bmatrix} F_{11} + G_1K_1 & F_{12} + G_1K_2 \\ 0 & F_{22} \end{bmatrix} \tag{6.12}$$

so that the pair $(F + GK, G)$ is still in standard reachability form, since (by Proposition 6.1.1) the subsystem $(F_{11} + G_1K_1, G_1)$ is reachable for every K_1. Thus, referring to an arbitrary basis for the state space, the structure of the Jordan form and the spectrum of the non-reachable system do not change, independently of the applied feedback law. ■

As a consequence of Proposition 6.1.2, to analyse the effects of state feedback on the eigenvalues of the system it is sufficient to analyse what happens to the reachable subspace. We will see that this analysis is simpler if we choose a suitable basis for the state space, so that the matrices F and G have suitable canonical structures.

6.2 Invariants and canonical forms

Up to now we used the terms *canonical* and *invariant* at a semi-intuitive level: we indeed proved that the transfer matrix of the system is "invariant" with respect to a change of basis in the state space and we used the Jordan "canonical form" to analyse the elementary modes of the system. Up to now, however, we did not discuss the precise meaning of these terms in Algebra. Since invariants and canonical forms will be used frequently in this chapter, it is convenient to formalise their definitions and illustrate them with some examples that are relevant to the study of linear systems.

Let $\mathcal{S}$ be a generic set. An *equivalence relation* $\sim$ in $\mathcal{S}$ is a relation that satisfies three properties: the relation is *reflexive*, *symmetric* and *transitive*, i.e., it satisfies for every $a, b, c \in \mathcal{S}$

$$a \sim a, \quad a \sim b \Rightarrow b \sim a, \quad a \sim b \text{ e } b \sim c \Rightarrow a \sim c. \tag{6.13}$$

An equivalence relation induces a partition of the set $\mathcal{S}$ into *equivalence classes*, characterised by the fact that all the elements within a given class are equivalent to each other, while elements that belong to different classes are (obviously) not equivalent. Given a generic $a \in \mathcal{S}$ we will denote by $[a]$ the equivalence class containing a. Notice that a given equivalence class contains in general more than one element, and thus it may be indicated with different symbols. It is important thus to remember that, despite having different names, this object is nonetheless the same.

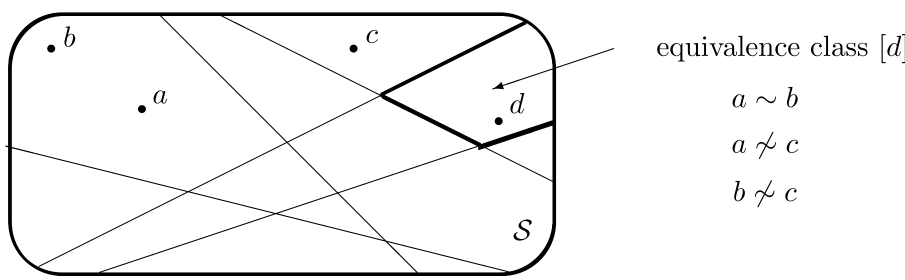


Figure 6.2.1

Equivalence classes are also called *quotient classes*. The set of all the equivalence classes induced by the equivalence relation $\sim$ over the generic set S is denoted by $\mathcal{S}/\sim$. The map

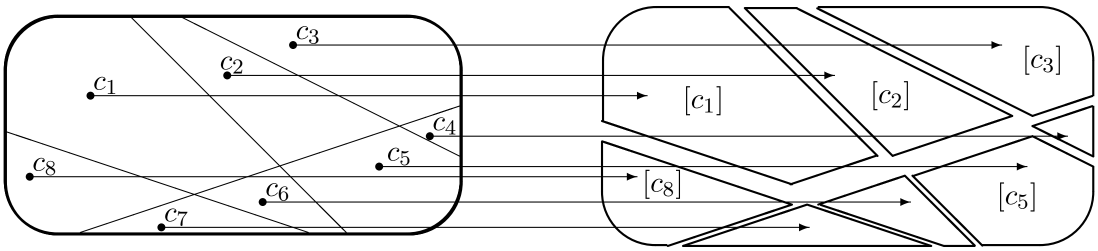
$$\mathcal{P}: \ \mathcal{S} \to \mathcal{S}/\sim : \ a \mapsto [a]$$

that associates to a generic element of S its equivalence class is instead called *projection map*, and it is obviously a surjective map.

Example 6.2.1 [Similarity among matrices] If $\mathbb{F}$ is a field, then introduce over the set $\mathcal{S} = \mathbb{F}^{n \times n}$ the similarity relation $\sim$ defined as follows: a matrix $A \in \mathcal{S}$ is similar to a matrix $B \in \mathcal{S}$ if there exists an invertible matrix $T \in \mathcal{S}$ such that $B = T^{-1}AT$.

It is immediate to verify that $\sim$ is reflexive, symmetric and transitive, so that it is an equivalence relation. Given a generic matrix A, the equivalence class $[A]$ contains all (and only) the matrices that can be written as $T^{-1}AT$ with T invertible. In other words, if A represents the linear transformation $\mathcal{A}: V \to V$ with respect to a specific n-dimensional basis of V then the equivalence class $[A]$ contains all and only the matrix representations of the linear transformation $\mathcal{A}$ for all possible bases of V.

Definition 6.2.1 [Canonical forms] *A set $\mathcal{C}$ of canonical forms for an equivalence relation $\sim$ over a set $\mathcal{S}$ is a subset of $\mathcal{S}$ for which every equivalence class of $\mathcal{S}/\sim$ contains only one element of $\mathcal{C}$. This means that if we limit the projection map $\mathcal{P}: \mathcal{S} \to \mathcal{S}/\sim$ to be applicable to the subset $\mathcal{C}$ then this map is bijective.*



$\mathcal{C} = \{c_1, c_2, \ldots, c_8\} \subset \mathcal{S}$ $\qquad$ $\mathcal{S} \underset{\mathcal{P}}{\longrightarrow} \mathcal{S}/\sim$

Example 6.2.2 Let $\mathbb{F} = \mathbb{C}$. A set of canonical forms for the similarity between matrices in $\mathbb{C}^{n\times n}$, defined in Example 6.2.1, is the set of all the Jordan canonical forms, identified with respect to the order of their mini-blocks. Indeed every complex-valued matrix of dimension $n \times n$ is similar to one and only one (neglecting the order of the mini-blocks) matrix in Jordan form. Another set of canonical forms for the similarity relation is the set of rational canonical forms.

Definition 6.2.2 [Invariants and complete invariants] *Let $\mathcal{S}$ be a generic set and $\sim$ a generic equivalence relation on $\mathcal{S}$. A function $f : \mathcal{S} \to \mathcal{T}$ is said to be an* invariant of $\mathcal{S}$ with respect to $\sim$ *if*

$$a \sim b \;\Rightarrow\; f(a) = f(b),\ \forall a, b \in \mathcal{S}.$$

The same function is said to be a complete invariant of $\mathcal{S}$ with respect to $\sim$ *if*

$$a \sim b \;\Leftrightarrow\; f(a) = f(b),\ \forall a, b \in \mathcal{S}.$$

Thus a function f is an invariant if it is constant on every class of $\mathcal{S}/\sim$, and is a complete invariant if it assumes different values on different classes[1].

Definition 6.2.3 [Complete set of invariants] *A set of functions $\{f_k : \mathcal{S} \to \mathcal{T}_k, k = 1, 2, \ldots, q\}$ is said to be a* complete set of invariants *(with respect to the equivalence relation $\sim$) if every function f_k is an invariant of $\mathcal{S}$ with respect to $\sim$ and if*

$$\left.\begin{array}{ccc} f_1(a) & = & f_1(b) \\ f_2(a) & = & f_2(b) \\ \ldots & \ldots & \ldots \\ f_q(a) & = & f_q(b) \end{array}\right\} \quad \Rightarrow \quad a \sim b.$$

Example 6.2.3 Let $\mathcal{S} = \mathbb{R}^{n\times n}$ and $\sim$ be the similarity between matrices in $\mathbb{C}^{n\times n}$. The function

$$f : \mathbb{R}^{n\times n} \to \mathbb{R}[s] : F \mapsto \Delta_F(s)$$

that associates to a matrix its characteristic polynomial is clearly an invariant. At the same time it is not a complete invariant, since there are matrices that are not similar but share the same characteristic polynomial.
Also the function

$$f : \mathbb{R}^{n\times n} \to \mathbb{R}[s] : F \mapsto \psi_F(s)$$

[1] Notice that if f is an invariant then the image of a, i.e., $f(a)$, is also called an invariant with respect to $\sim$, meaning with this that $f(a)$ and $f(b)$ coincide for every $b \sim a$.

that associates to a matrix its minimal annihilating polynomial is an (incomplete) invariant.

The map g that associates to a generic matrix F the set $\{\psi_1(s), \dots, \psi_c(s)\}$ of its invariant polynomials is instead a complete invariant. Indeed, as shown in Section A.10, two matrices are similar if and only if they have the same invariant polynomials.

Exercise 6.2.1 Prove that the trace of a matrix (i.e., the sum of its diagonal elements) is an invariant for the similarity between matrices. In other words, prove that similarity transformations do not change traces.

♯ *Hint: as a first step prove that for any pair of $n \times n$ matrices A and B the traces of AB and BA are equal. Then the trace of $T^{-1}AT$ coincides with the trace of $(AT)T^{-1}$.*

Example 6.2.4 [Algebraic equivalence of LTI systems] The algebraic equivalence between linear time invariant systems is an equivalence relation over the set $\mathcal{S} = \mathbb{R}^{n\times n} \times \mathbb{R}^{n\times m} \times \mathbb{R}^{p\times n}$ of matrix triplets describing linear time invariant systems of dimension n and with m inputs and p outputs.

An equivalence class of $\mathcal{S}$ with respect to this equivalence relation contains all the matrix representations of the same dynamical system for all the possible bases of the state space. Thus, if one wants to analyse some properties that are invariant with respect to a change of the basis of the state space, one may consider indifferently one of the various representations within a given equivalence class. Nonetheless choosing an appropriate basis leads to matrix structures that are in some sense easier to be used, in the sense that they visually highlight some of the properties that one wants to analyse. The purpose of the canonical forms we will introduce in this chapter and in the next one are meant to provide, within each equivalence class, specific triplets whose structure allows to easily design feedback laws and state observers.

Some of the invariants for the previous algebraic equivalence relation are then

i) the rank of the reachability matrix and, in the case of discrete-time systems, the rank of the reachability matrices in k steps for any k;
ii) considering the subset of $\mathcal{S}$ of the fully reachable systems, the reachability index r;
iii) the transfer matrix;
iv) the dimension of the controllable subspace in k steps for the case of discrete-time systems.

Here we prove only the point (iv), since the previous points have either been proved before or are trivial. Let then $\Sigma_1 = (F_1, G_1, H_1)$ and $\Sigma_2 = (F_2, G_2, H_2)$ be two discrete-time systems that are algebraically equivalent through the invertible transformation matrix T, i.e., are such that

$$F_2 = T^{-1}F_1T,\ G_2 = T^{-1}G_1,\ H_2 = H_1T.$$

Letting $\mathcal{R}_k^{(i)}$ and $(X^{(i)})_k^C$ be the reachability matrix and the controllable subspace, respectively, in k steps of the system $\Sigma_i,\ i = 1, 2$, it holds

$$\mathcal{R}_k^{(2)} = T^{-1}\mathcal{R}_k^{(1)}$$

and

$$\begin{aligned}
(X^{(2)})_k^C &= \{\mathbf{x} \in \mathbb{R}^n : F_2^k\mathbf{x} \in \mathrm{Im}\Big(\mathcal{R}_k^{(2)}\Big)\} = \{\mathbf{x} \in \mathbb{R}^n : T^{-1}F_1^kT\mathbf{x} \in T^{-1}\mathrm{Im}\Big(\mathcal{R}_k^{(1)}\Big)\} \\
&= \{\mathbf{x} \in \mathbb{R}^n : F_1^kT\mathbf{x} \in \mathrm{Im}\Big(\mathcal{R}_k^{(1)}\Big)\} = \{\mathbf{x} \in \mathbb{R}^n : T\mathbf{x} \in (X^{(1)})_k^C\} \\
&= \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \in T^{-1}(X^{(1)})_k^C\} = T^{-1}(X^{(1)})_k^C.
\end{aligned}$$

Thus $(X^{(1)})_k^C$ and $(X^{(2)})_k^C$ have the same dimension and, obviously, they contain n-tuples that represent the same vectors referred to two different bases.

6.3 Controllable canonical form

As said at the end of Section 6.1, when studying state feedback it is useful to consider reachable systems (F, G) that are expressed in particular canonical forms. For single-input systems this canonical form is the so-called *controllable canonical form*, mathematically expressed as

$$F_c = \begin{bmatrix} 0 & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \dots & -\alpha_{n-1} \end{bmatrix}, \quad \mathbf{g}_c = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}. \tag{6.14}$$

Lemma 6.3.1 *Independently of $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$ in (6.14),*

(i) the pair $(F_c, \mathbf{g}_c)$ is reachable;

(ii) the characteristic polynomial of F_c is

$$\Delta_{F_c}(s) = \alpha_0 + \alpha_1 s + \dots + \alpha_{n-1} s^{n-1} + s^n; \tag{6.15}$$

(iii) the elements of the n-th column of the matrix $\operatorname{adj}(sI_n - F_c)$ are the terms 1, s, s^2, $\dots$, s^{n-1}.

Proof (i) The reachability matrix of the system $(F_c, \mathbf{g}_c)$ is

$$\mathcal{R}_c = [\,\mathbf{g}_c \quad F_c\mathbf{g}_c \quad F_c^2\mathbf{g}_c \quad \dots \quad F_c^{n-1}\mathbf{g}_c\,] = \begin{bmatrix} & & & & 1 \\ & & & \dots & * \\ & & 1 & * & * \\ & 1 & * & * & * \\ 1 & * & * & * & * \end{bmatrix}.$$

The determinant of this matrix is either $+1$ or -1, thus the system is reachable.
(ii) See Proposition A.10.2.
(iii) The algebraic complement of the (n, j)-th entry of the matrix $sI_n - F_c$ is

$$(-1)^{n+j} \det \left[\begin{array}{cccc|cccc} s & -1 & & & & & & \\ & \ddots & \ddots & & & & & \\ & & s & -1 & & & & \\ & & & s & & & & \\ \hline & & & & -1 & & & \\ & & & & s & -1 & & \\ & & & & & \ddots & \ddots & \\ & & & & & & s & -1 \end{array} \right] = (-1)^{n+j} s^{j-1} (-1)^{n-j},$$

thus the (j, n)-th element of the adjoint matrix is s^{j-1}. ■

When discussing state-feedback problems for reachable single-input systems $(F, \mathbf{g})$ we will always refer to the canonical form (6.14). Indeed:

Proposition 6.3.2 *A single-input system $\Sigma = (F, \mathbf{g}, H)$ is reachable if and only if it is algebraically equivalent to a system $\Sigma_c = (F_c, \mathbf{g}_c, H_c)$ with $(F_c, \mathbf{g}_c)$ in controllable canonical form.*

Proof If the n-dimensional system $(F, \mathbf{g}, H)$ is reachable, then the column vectors of the reachability matrix $\mathcal{R} = [\,\mathbf{g} \quad F\mathbf{g} \quad F^2\mathbf{g} \quad \dots \quad F^{n-1}\mathbf{g}\,]$ are linearly independent and constitute a cyclic basis for $\mathbb{R}^n$. To obtain a system with the form (6.14), we can build an alternative basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ of $\mathbb{R}^n$ starting from the former cyclic basis. To this aim, assume that the characteristic polynomial of F is $\Delta_F(s) = \alpha_0 + \alpha_1 s + \ldots + \alpha_{n-1}s^{n-1} + s^n$. Then introduce the n linearly independent columns

$$\begin{aligned}
\mathbf{v}_1 &= F^{n-1}\mathbf{g} + \alpha_{n-1}F^{n-2}\mathbf{g} + \ldots + \alpha_2 F\mathbf{g} + \alpha_1\mathbf{g} \\
\mathbf{v}_2 &= F^{n-2}\mathbf{g} + \alpha_{n-1}F^{n-3}\mathbf{g} + \ldots + \alpha_2\mathbf{g} \\
\dots & \quad \dots\dots \\
\mathbf{v}_{n-2} &= F^2\mathbf{g} + \alpha_{n-1}F\mathbf{g} + \alpha_{n-2}\mathbf{g} \\
\mathbf{v}_{n-1} &= F\mathbf{g} \;+\; \alpha_{n-1}\mathbf{g} \\
\mathbf{v}_n &= \mathbf{g}
\end{aligned} \tag{6.16}$$

For $i = 2, 3, ...n$ it holds that

$$F\mathbf{v}_i = \mathbf{v}_{i-1} - \alpha_{i-1}\mathbf{v}_n.$$

For $i = 1$, instead, by the Cayley Hamilton theorem, it holds that

$$F\mathbf{v}_1 = F^n\mathbf{g} + \alpha_{n-1}F^{n-1}\mathbf{g} + \ldots + \alpha_2 F^2\mathbf{g} + \alpha_1 F\mathbf{g} = -\alpha_0\mathbf{g}.$$

The basis (11.13) can be obtained from the original basis of $\Sigma = (F, \mathbf{g}, H)$ by exploiting the transformation matrix

$$T = [\,\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_n\,] = [\,\mathbf{g} \quad F\mathbf{g} \quad \dots \quad F^{n-1}\mathbf{g}\,] \begin{bmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_{n-1} & 1 \\ \alpha_2 & \dots & \alpha_{n-1} & 1 & \\ \vdots & \dots & \dots & & \\ \alpha_{n-1} & 1 & & & \\ 1 & & & & \end{bmatrix}. \tag{6.17}$$

It is then immediate to verify that, with respect to the basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$, the system takes the controllable canonical form (6.14), i.e.,

$$T^{-1}FT = F_c \quad T^{-1}\mathbf{g} = \mathbf{g}_c. \tag{6.18}$$

Viceversa, suppose that there is a change of basis for which the pair $(F, \mathbf{g})$ reduces to controllable canonical form. Then there exists a transformation matrix T for which the relations (6.18) hold. Since the pair $(F_c, \mathbf{g}_c)$ is reachable, also the pair $(F, \mathbf{g})$ must be reachable, since the rank of reachability matrices does not change when changing the basis of X. ■

Exercise 6.3.1 [Recap] Let $F \in \mathbb{R}^{n\times n}$ and $\mathbf{g} \in \mathbb{R}^n$. Then:

1. The following facts are equivalent:
   (a) F has a cyclic vector (i.e., F is a cyclic matrix);

   (b) F is similar to a companion matrix;
   (c) the Jordan form of F has a single mini-block for each eigenvalue;
   (d) the characteristic polynomial and the minimal polynomial of F coincide;
   (e) the greatest common divisor of the elements of $\mathrm{adj}(zI - F)$ is 1;
   (f) $\mathrm{adj}(zI - F) \neq 0, \ \forall z \in \mathbb{C}$.

2. The following facts are equivalent:
   (a) $(F, \mathbf{g})$ is reachable;
   (b) $(F, \mathbf{g})$ is algebraically equivalent to a system $(F_c, \mathbf{g}_c)$ in controllable canonical form ;
   (c) $(F, \mathbf{g})$ is algebraically equivalent to a system $(F_J, \mathbf{g}_J)$ as in (4.85) with F_J in Jordan form;
   (d) F is a cyclic matrix and $\mathbf{g}$ is a cyclic vector for F;
   (e) the matrix $[\, zI - F \quad \mathbf{g}\,]$ has full rank for every $z \in \mathbb{C}$.

Remark If the pair $(F, \mathbf{g})$ is reachable, then to obtain the matrix F_c of the corresponding controllable canonical form it is sufficient to know the coefficients of the characteristic polynomial $\Delta_F(s)$, so that it is not necessary to determine also the transformation matrix T. We remark, however, that the controllable canonical form $(F_c, \mathbf{g}_c)$ obtained in this way is algebraically equivalent to the pair $(F, \mathbf{g})$ *only if* $(F, \mathbf{g})$ is reachable.
If the problem to be solved requires also the knowledge of the transformation matrix T then this matrix can be computed starting from the reachability matrices $\mathcal{R}$ and $\mathcal{R}_c$ relative to $(F, \mathbf{g})$ and $(F_c, \mathbf{g}_c)$, respectively. Indeed, by exploiting $\mathcal{R}_c = T^{-1}\mathcal{R}$ we can directly obtain

$$T = \mathcal{R}\mathcal{R}_c^{-1}, \tag{6.19}$$

where the explicit expression for $\mathcal{R}_c^{-1}$ can be obtained by comparing (6.17) and (6.19):

$$\mathcal{R}_c^{-1} = \begin{bmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_{n-1} & 1 \\ \alpha_2 & \dots & \alpha_{n-1} & 1 & \\ \vdots & \dots & \dots & & \\ \alpha_{n-1} & 1 & & & \\ 1 & & & & \end{bmatrix} \tag{6.20}$$

When a single-input single-output system is in controllable canonical form then the determination of its transfer matrix is particularly simple. Indeed from the matrices

$$F_c = \begin{bmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ -\alpha_0 & -\alpha_1 & \dots & -\alpha_{n-1} \end{bmatrix}, \quad \mathbf{g}_c = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, \quad H_c = [\, \beta_0 \quad \beta_1 \quad \dots \quad \beta_{n-1}\,]$$

and considering Lemma 6.3.1 it is possible to immediately obtain

$$W(s) \;=\; H_c(sI_n - F_c)^{-1}\mathbf{g}_c = [\, \beta_0 \quad \beta_1 \quad \dots \quad \beta_{n-1}\,] \, \frac{\mathrm{adj}(sI_n - F_c)}{\det(sI_n - F_c)} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \beta_0 & \beta_1 & \dots & \beta_{n-1} \end{bmatrix} \begin{bmatrix} 1 \\ s \\ \vdots \\ s^{n-1} \end{bmatrix} / \det(sI_n - F_c)$$

$$= \frac{\beta_0 + \beta_1 s + \ldots + \beta_{n-1}s^{n-1}}{\alpha_0 + \alpha_1 s + \ldots + \alpha_{n-1}s^{n-1} + s^n}. \tag{6.21}$$

Numerator and denominator of the transfer function of the system are associated to the elements of H_c and to the elements of the last row of F_c, respectively. Notice that any rational function $W(s)$ admits an infinite number of different representations as the ratio of two polynomials, each obtainable from the coprime representation by multiplying the numerator and the denominator by the same non null polynomial factor. Importantly, in (6.21) the polynomials $\sum_i \beta_i s^i$ and $\sum_i \alpha_i s^i$ may not be coprime. This means that (6.21) is not necessarily an irreducible representation of $W(s)$.

We will see in the chapter devoted to the theory of realization of linear systems that the fact that the polynomials $H_c\mathrm{adj}(sI_n - F_c)\mathbf{g}_c$ and $\det(sI_n - F_c)$ are coprime is actually a necessary and sufficient condition for the minimality of $(F_c, \mathbf{g}_c, H_c)$.

6.4 Allocation of the eigenvalues: case $m = 1$

Consider a reachable system $(F, \mathbf{g})$ and a change of basis $\bar{\mathbf{x}} = T^{-1}\mathbf{x}$ over the state space that reduces the original system to its controllable canonical form (6.14). Then if we adopt a control law of the type

$$u(t) = K_c\bar{\mathbf{x}}(t) + v(t),$$

with

$$K_c = \begin{bmatrix} k_{c,0} & k_{c,1} & \dots & k_{c,n-1} \end{bmatrix}$$

then we obtain the state update law with respect to the new basis

$$\bar{\mathbf{x}}(t+1) = (F_c + \mathbf{g}_c K_c)\bar{\mathbf{x}}(t) + \mathbf{g}_c v(t) \tag{6.22}$$

with $F_c + \mathbf{g}_c K_c$ in companion form:

$$F_c + \mathbf{g}_c K_c = \begin{bmatrix} 0 & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ -\alpha_0 + k_{c,0} & -\alpha_1 + k_{c,1} & -\alpha_2 + k_{c,2} & \dots & -\alpha_{n-1} + k_{c,n-1} \end{bmatrix}. \tag{6.23}$$

Notice that the characteristic polynomial of the closed-loop system is given by

$$\Delta_{F_c + \mathbf{g}_c K_c}(s) = (\alpha_0 - k_{c,0}) + (\alpha_1 - k_{c,1})s + (\alpha_{n-1} - k_{c,n-1})s^{n-1} + s^n.$$

This means that this monic polynomial can be equated to any monic polynomial of the same degree

$$p(s) = p_0 + p_1 s + p_2 s^2 + \dots p_{n-1} s^{n-1} + s^n \tag{6.24}$$

by properly choosing the coefficients of the feedback matrix as

$$k_{c,i} = \alpha_i - p_i \qquad i = 0, 1, \dots, n-1. \tag{6.25}$$

This means that K_c can be chosen so that the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of the state update matrix $F_c + \mathbf{g}_c K_c$ can be arbitrarily set (with the constraint that complex eigenvalues must be present in complex-conjugate pairs). It will then be enough to set $p(s) = \prod_{i=1}^{n}(s - \lambda_i)$ in (6.24).

The matrix K to be used in the control law for the system in its original representation (i.e., $(F, \mathbf{g})$) that needs to be chosen to ensure that the closed loop system $F + \mathbf{g}K$ has the aforementioned eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ above can be computed directly from T and K_c, since

$$F_c + \mathbf{g}_c K_c = T^{-1} F T + T^{-1} \mathbf{g} K_c T^{-1} T = T^{-1} \left[F + \mathbf{g}(K_c T^{-1}) \right] T. \tag{6.26}$$

Since similar matrices have the same spectrum, the eigenvalues of $F_c + \mathbf{g}_c K_c$ are equal to the ones of $F + \mathbf{g}(K_c T^{-1})$, so it will be enough to choose

$$K = K_c T^{-1}.$$

Proceeding in this way we constructively proved the following claim:

Proposition 6.4.1 [Allocation of the eigenvalues, case $m = 1$] *Let $(F, \mathbf{g}, H)$ be an n-dimensional reachable system. Then fore every monic polynomial of order n*

$$p(s) = p_0 + p_1 s + p_2 s^2 + \dots p_{n-1} s^{n-1} + s^n \in \mathbb{R}[s]$$

there exists a matrix $K \in \mathbb{R}^{1 \times n}$ such that the characteristic polynomial of $F + \mathbf{g}K$ is $p(s)$. ∎

Example 6.4.1 Consider the system $\dot{\mathbf{x}}(t) = F\mathbf{x}(t) + G\mathbf{u}(t)$ for which

$$F = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}. \tag{6.27}$$

The reachability matrix of the system is

$$\mathcal{R} = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

and has rank 3. The characteristic polynomial of F is instead

$$\Delta_F(s) = \alpha_0 + \alpha_1 s + \alpha_2 s^2 + s^3 = 1 - s - s^2 + s^3.$$

Thus $(F, \mathbf{g})$ is fully reachable and its controllable canonical form can be obtained immediately from the characteristic polynomial of F as

$$F_c = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}, \quad \mathbf{g}_c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The transformation matrix that reduces $(F, \mathbf{g})$ to its controllable canonical form is thus $T = \mathcal{R}\mathcal{R}_c^{-1}$ and its inverse is

$$T^{-1} = \mathcal{R}_c\mathcal{R}^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Suppose one wants to allocate the eigenvalues of the closed loop system $(F + \mathbf{g}K, \mathbf{g})$ in $-2, -2, -1$ or, equivalently, to impose that the characteristic polynomial of the matrix $F + \mathbf{g}K$ is

$$p(s) = (s+2)^2(s+1) = s^3 + 5s^2 + 8s + 4.$$

To this aim we consider first the very same problem for the controllable canonical form, i.e., we start by imposing that the characteristic polynomial of

$$F_c + \mathbf{g}_c K_c = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 + k_{c,0} & 1 + k_{c,1} & 1 + k_{c,2} \end{bmatrix}$$

coincides with $p(s)$ defined above. Then we must solve the system of linear equations

$$\begin{aligned} -4 &= -1 + k_{c,0} \\ -8 &= 1 + k_{c,1} \\ -5 &= 1 + k_{c,2}. \end{aligned}$$

From the matrix $K_c = [\,k_{c,0} \quad k_{c,1} \quad k_{c,2}\,] = [\,-3 \quad -9 \quad -6\,]$ obtained in this way we obtain

$$K = K_c T^{-1} = [\,-9 \quad -6 \quad 3\,].$$

The procedure described above shows that the row vector $K = [\,k_0 \quad k_1 \quad \dots \quad k_{n-1}\,]$ is the solution of the linear system in the unknowns k_i

$$[\,k_0 \quad k_1 \quad \dots \quad k_{n-1}\,]\,T = [\,\alpha_0 - p_0 \quad \alpha_1 - p_1 \quad \dots \quad \alpha_{n-1} - p_{n-1}\,] \tag{6.28}$$

The entries in the vector of the known coefficients are the differences between the α_i's and the p_i's, that represent the coefficients of the characteristic polynomials of F and of $F + \mathbf{g}K$, respectively.

Since the invertibility of T guarantees the existence and uniqueness of the solution, there is an alternative procedure for determining K: it consists in directly imposing that the coefficients of the polynomial $\det(sI - F - \mathbf{g}K)$, that are functions of the parameters k_i, coincide with the coefficients of $p(s)$. In this way one obtains another system of linear equations in the unknowns k_i that is equivalent to the system defined in (7.28).

Example 6.4.1 - (Continue) Alternatively to the previous method, consider then the equation

$$\det(sI_n - F - \mathbf{g}K) = p(s)$$

that has as its unknowns the components of K, i.e.,

$$\begin{aligned} &\det\left(\begin{bmatrix} s-1 & -2 & 0 \\ 0 & s & -1 \\ 0 & -1 & s \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} [\,k_0 \quad k_1 \quad k_2\,] \right) \\ &= s^3 + (-k_0 - k_2 - 1)s^2 + (-k_1 + k_2 - 1)s + (-k_0 + k_1 + 1) \\ &= s^3 + 5s^2 + 8s + 4. \end{aligned}$$

By imposing that the polynomials in the second and third rows coincide we obtain the algebraic system

$$\begin{array}{rcrcr} k_0 & + & k_2 & = & -6 \\ -k_1 & + & k_2 & = & 9 \\ k_0 & - & k_1 & = & -3 \end{array} \tag{6.29}$$

whose solution is $K = [\,k_0 \quad k_1 \quad k_2\,] = [\,-9 \quad -6 \quad 3\,]$.

For a single-input single-output reachable system, it is very simple to analyse the effect of a state feedback law on its transfer function. Indeed transfer functions do not depend on the basis chosen for representing the system. Thus we may assume without loss of generality that both the open loop and the closed loop systems are represented in their respective controllable canonical forms. Given the discussion above, taking the open loop system $\Sigma = (F_c, \mathbf{g}_c, H_c)$ and closing the loop with the matrix $K_c = [\,k_{c,0} \quad k_{c,1} \quad \ldots k_{c,n-1}\,]$ leads to the closed loop system $\Sigma^{(K_c)}$ with transfer function

$$\begin{aligned} w_{K_c}(s) &= \frac{H_c \mathrm{adj}(sI_n - F_c - \mathbf{g}_c K_c)\mathbf{g}_c}{\det(sI_n - F_c - \mathbf{g}_c K_c)} \\ &= \frac{\beta_0 + \beta_1 s + \ldots + \beta_{n-1} s^{n-1}}{(\alpha_0 - k_{c,0}) + (\alpha_1 - k_{c,1})s + \ldots + (\alpha_{n-1} - k_{c,n-1})s^{n-1} + s^n}. \end{aligned} \tag{6.30}$$

It is immediate to notice that the introduction of the feedback modifies the poles of the original open loop transfer function (6.21). The zeros, instead, are unchanged and thus do not depend on the design parameters $k_{c,i}$ as soon as there are no cancellations between the numerator and denominator of (6.30) (or, viceversa, as far as the original cancellations between the numerator and denominator of (6.21) no longer appear in (6.30)).

Example 6.4.2 Consider again the controllable canonical form of the system of Example 6.4.1,

$$F_c = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}, \quad \mathbf{g}_c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Suppose that the output matrix is

$$H_c = [\,2 \quad -3 \quad 1\,]$$

so that

$$w(s) = \frac{H_c \mathrm{adj}(sI_n - F_c)\mathbf{g}_c}{\det(sI_n - F_c)} = \frac{s^2 - 3s + 2}{s^3 - s^2 - s + 1} = \frac{(s-1)(s-2)}{(s-1)^2(s+1)} = \frac{s-2}{(s-1)(s+1)}.$$

In this case $w(s)$ has two single poles in $+1$ and -1, and a single zero in 2. If we choose $K_c = [\,k_{c,0} \quad k_{c,1} \quad k_{c,2}\,]$ as feedback matrix, then

$$\begin{aligned} w_{K_c}(s) &= \frac{H_c \mathrm{adj}(sI_n - F_c - \mathbf{g}_c K_c)\mathbf{g}_c}{\det(sI_n - F_c - \mathbf{g}_c K_c)} \\ &= \frac{s^2 - 3s + 2}{s^3 - (1 + k_{c,2})s^2 - (1 + k_{c,1})s - (-1 + k_{c,0})} \\ &= \frac{(s-1)(s-2)}{(s-\lambda_1)(s-\lambda_2)(s-\lambda_3)} \end{aligned}$$

where $\lambda_1, \lambda_2, \lambda_3$ are the eigenvalues (that can be arbitrarily set) of $F_c + \mathbf{g}_c K_c$. It is clear that the zeros of the transfer function $w_{K_c}(s)$ (positioned in 1 and 2) remain unchanged as $\lambda_1, \lambda_2, \lambda_3$ vary, as long as these λ_i's do not lead to numerator-denominator cancellations. Notice also that, in the considered case, after having applied a state feedback to the system in general the cancellation of the factor $(s-1)$ between the numerator and denominator of $w(s)$ does not take place anymore.

To summarise, the allocation technique described above essentially requires to compute the companion form of the matrix F and the determination of the coefficients of $\Delta_F(s)$ and of $\Delta_{F+\mathbf{g}K}(s)$. We will now describe two alternative methods for allocating the eigenvalues: the first one is valid in general, while the second can be applied only when the matrix F is diagonal.

Proposition 6.4.2 [Ackermann's formula] *Assume that $(F, \mathbf{g})$ is reachable, with reachability matrix $\mathcal{R}$. Then the characteristic polynomial of the closed loop system $(F + \mathbf{g}K, \mathbf{g})$ is equal to the monic polynomial*

$$p(s) = p_0 + p_1 s + \ldots + p_{n-1}s^{n-1} + s^n$$

as far as

$$K = -\mathbf{e}_n^T \mathcal{R}^{-1} p(F), \tag{6.31}$$

with $\mathbf{e}_n$ being the n-th vector of the canonical basis of $\mathbb{R}^n$.

Proof Suppose that the system is in controllable canonical form, i.e.,

$$F_c = \begin{bmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ -\alpha_0 & -\alpha_1 & \ldots & -\alpha_{n-1} \end{bmatrix}, \ \mathbf{g}_c = \mathbf{e}_n.$$

Notice then that, due to (6.20), the last row of the matrix $\mathcal{R}_c^{-1}$ coincides with $\mathbf{e}_1^T$. Moreover,

$$\mathbf{e}_1^T F_c^i = \mathbf{e}_{i+1}^T, \ i = 0, 1, \ldots, n-1.$$

Thus

$$\begin{aligned} F_c \ &+\ \mathbf{g}_c\left(-\mathbf{e}_n^T \mathcal{R}_c^{-1} p(F_c)\right) = F_c - \mathbf{e}_n \mathbf{e}_1^T \left(\sum_{i=0}^{n-1} p_i F_c^i + F_c^n\right) \\ &= \ F_c - \mathbf{e}_n \mathbf{e}_1^T \left(\sum_{i=0}^{n-1} p_i F_c^i - \sum_{i=0}^{n-1} \alpha_i F_c^i\right) = F_c + \mathbf{e}_n \left(\sum_{i=0}^{n-1} (\alpha_i - p_i)\mathbf{e}_1^T F_c^i\right) \\ &= \ F_c + \mathbf{e}_n \left(\sum_{i=0}^{n-1} (\alpha_i - p_i)\mathbf{e}_{i+1}^T\right) = \begin{bmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ -p_0 & -p_1 & \ldots & -p_{n-1} \end{bmatrix} \end{aligned}$$

and hence $F_c + \mathbf{g}_c K_c$ has $p(s)$ as its characteristic polynomial.
In the general case if T is the transformation matrix that reduces the original system $(F, \mathbf{g})$ in its controllable canonical form, then $p(s)$ is also the characteristic polynomial of

$$TF_cT^{-1} + T\mathbf{g}_c[K_cT^{-1}] \ = \ F + \mathbf{g}\left[-\mathbf{e}_n^T \mathcal{R}_c^{-1} T^{-1} T p(F_c) T^{-1}\right]$$

$$\begin{aligned} &= F + \mathbf{g}\left[-\mathbf{e}_n^T (T\mathcal{R}_c)^{-1} p(TF_cT^{-1})\right] \\ &= F + \mathbf{g}\left[-\mathbf{e}_n^T \mathcal{R}^{-1} p(F)\right] = F + \mathbf{g}K \qquad \blacksquare \end{aligned}$$

Assume that $(F, \mathbf{g})$ is a reachable system and F is a diagonal matrix, with (necessarily distinct) eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. In this case one can express the elements of the feedback matrix

$$K = [\, k_1 \quad k_2 \quad \ldots \quad k_n \,]$$

as functions of the differences between the eigenvalues $\mu_1, \mu_2, \ldots, \mu_n$ of the closed loop system $F + \mathbf{g}K$ and the original eigenvalues of F.

Recall then that if two matrices A and B have dimension $n \times m$ and $m \times n$ then[2] $\det(I_n + AB) = \det(I_m + BA)$. This implies that the relationship between the polynomials $\Delta_{F+\mathbf{g}K}(s)$ and $\Delta_F(s)$ is

$$\begin{aligned} \Delta_{F+\mathbf{g}K}(s) &= \det\left(sI_n - F - \mathbf{g}K\right) \\ &= \det\left[(sI_n - F)(I_n - (sI_n - F)^{-1}\mathbf{g}K)\right] \\ &= \Delta_F(s) \det\left(I_n - (sI_n - F)^{-1}\mathbf{g}K\right) \\ &= \Delta_F(s)\left(1 - K(sI_n - F)^{-1}\mathbf{g}\right). \end{aligned} \tag{6.32}$$

Since

$$\begin{aligned} 1 \; - \; & K(sI_n - F)^{-1}\mathbf{g} \\ = \; & 1 - [\, k_1 \quad k_2 \quad \ldots \quad k_n \,] \begin{bmatrix} (s-\lambda_1)^{-1} & & & \\ & (s-\lambda_2)^{-1} & & \\ & & \ddots & \\ & & & (s-\lambda_n)^{-1} \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{bmatrix} \\ = \; & 1 - \sum_{i=1}^{n} \frac{k_i g_i}{s - \lambda_i}, \end{aligned}$$

and since

$$\frac{\Delta_{F+\mathbf{g}K}(s)}{\Delta_F(s)} = \frac{\prod_{i=1}^n (s-\mu_i)}{\prod_{i=1}^n (s-\lambda_i)}$$

from (6.32) it is possible to obtain

$$\frac{\prod_{i=1}^n (s-\mu_i)}{\prod_{i=1}^n (s-\lambda_i)} = 1 - \sum_{i=1}^{n} \frac{k_i g_i}{s - \lambda_i}. \tag{6.33}$$

[2] To this aim it is sufficient to remember that the determinant of the product of two square matrices coincides with the product of the corresponding determinants, so that

$$\det(I_n + AB) = \det\left(\begin{bmatrix} I_n & A \\ 0 & I_m \end{bmatrix}\begin{bmatrix} I_n & -A \\ B & I_m \end{bmatrix}\right) = \det\left(\begin{bmatrix} I_n & -A \\ B & I_m \end{bmatrix}\begin{bmatrix} I_n & A \\ 0 & I_m \end{bmatrix}\right) = \det(I_m + BA).$$

By identifying the residuals at the pole λ_j for both members of (6.33), we obtain

$$\frac{\displaystyle\prod_{i=1}^{n}(\lambda_j - \mu_i)}{\displaystyle\prod_{\substack{i=1\\ i\neq j}}^{n}(\lambda_j - \lambda_i)} = -k_j g_j.$$

Keeping into account that all the entries of $\mathbf{g}$ are different from zero, we prove the following claim:

Proposition 6.4.3 [Mayne-Murdoch formula] *Let $(F, \mathbf{g})$ be reachable and let F be diagonal with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Given any n-tuple $\mu_1, \mu_2, \ldots, \mu_n$ of complex numbers, the matrix $K = [\, k_1 \quad k_2 \quad \ldots \quad k_n \,]$ with*

$$k_j = -\frac{1}{g_j}\frac{\displaystyle\prod_{i=1}^{n}(\lambda_j - \mu_i)}{\displaystyle\prod_{\substack{i=1\\ i\neq j}}^{n}(\lambda_j - \lambda_i)} \tag{6.34}$$

attributes to $F + \mathbf{g}K$ the eigenvalues $\mu_1, \mu_2, \ldots, \mu_n$. ■

An interesting consequence of (6.34) is that the gain of the closed loop grows with the distance $|\lambda_j - \mu_i|$ between the eigenvalues of the original system and of the closed loop one.

6.5 Allocation of the eigenvalues: case $m > 1$

Consider a reachable system $\Sigma = (F, G, H)$ of dimension n with more than one input. As for the case of single-input systems, we can prove that the eigenvalues of the matrix $F + GK$ of the closed loop system

$$\dot{\mathbf{x}}(t) = (F + GK)\mathbf{x}(t) + G\mathbf{v}(t)$$

can be arbitrarily placed in the complex plane by properly choosing K. The procedure that we will discuss here consists of two steps:

1. introduce a state feedback that makes the resulting closed loop system reachable from just one entry of $\mathbf{v}$;
2. apply one of the techniques proposed in the previous section to allocate the eigenvalues of the closed loop system obtained in the first step.

Note that, since the overall system $\Sigma = (F, G, H)$ is reachable, the matrix

$$\mathcal{R} = [\, \underbrace{\mathbf{g}_1 \;\; \mathbf{g}_2 \;\; \cdots \;\; \mathbf{g}_m}_{G} \;\; \underbrace{F\mathbf{g}_1 \;\; F\mathbf{g}_2 \;\; \cdots \;\; F\mathbf{g}_m}_{FG} \;\; \cdots \;\; \underbrace{F^{n-1}\mathbf{g}_1 \;\; F^{n-1}\mathbf{g}_2 \;\; \cdots \;\; F^{n-1}\mathbf{g}_m}_{F^{n-1}G} \,]$$

has rank n. If there exists a vector $\mathbf{g}_i$ such that the sub-matrix

$$[\,\mathbf{g}_i \quad F\mathbf{g}_i \quad \cdots \quad F^{n-1}\mathbf{g}_i\,] \tag{6.35}$$

has rank n, then the system is reachable from the i-th input. In this case we can skip the first step of the procedure described above: we can indeed refer to the pair $(F, \mathbf{g}_i)$, and construct a row matrix K_i such that $F + \mathbf{g}_i K_i$ has the desired eigenvalues. Since

$$F + \mathbf{g}_i K_i = F + [\,\mathbf{g}_1 \quad \cdots \quad \mathbf{g}_i \quad \cdots \quad \mathbf{g}_m\,] \begin{bmatrix} 0 \\ \vdots \\ K_i \\ \vdots \\ 0 \end{bmatrix}$$

it is sufficient to choose for the whole system (F, G) the $m \times n$-dimensional feedback matrix

$$\bar{K} := \begin{bmatrix} 0 \\ \vdots \\ K_i \\ \vdots \\ 0 \end{bmatrix}$$

whose rows are all null except for the i-th one, that will coincide with K_i.

If there is no vector $\mathbf{g}_i$ such that the matrix (6.35) has rank n, then the various systems $(F, \mathbf{g}_i)$ are not (entirely)completely) reachable for $i = 1, 2 \ldots, m$. In this case, to arbitrarily allocate the eigenvalues of the whole system it is necessary to introduce a feedback law that exploits more than one input.

The first step of the procedure, based on Heymann's lemma, requires to design a state feedback that makes the system reachable from a single input:

Proposition 6.5.1 [Heymann's lemma] *If (F, G) is reachable and if $\mathbf{g}_i$ is a non-null column of G, then there exists a matrix $M \in \mathbb{R}^{m \times n}$ such that $(F + GM, \mathbf{g}_i)$ is reachable.*

Proof Without loss of generality, assume $i = 1$. Choose then n linearly independent columns in $\mathcal{R}$ by following this procedure:

1. consider the sequence of vectors $F^h\mathbf{g}_1$ for $h = 0, 1, \ldots$, and stop at the first integer ν_1 for which $F^{\nu_1}\mathbf{g}_1$ is linearly dependent on the preceding vectors

$$\mathbf{g}_1, F\mathbf{g}_1, \ldots, F^{\nu_1 - 1}\mathbf{g}_1.$$

Notice that it must be $\nu_1 < n$, otherwise the pair $(F, \mathbf{g}_1)$ would be reachable. Consider then the sequence $F^h\mathbf{g}_2$ for $h = 0, 1, \ldots$, and stop at the first integer ν_2 for which $F^{\nu_2}\mathbf{g}_2$ is linearly dependent on the preceding vectors

$$\mathbf{g}_1, F\mathbf{g}_1, \ldots, F^{\nu_1 - 1}\mathbf{g}_1, \mathbf{g}_2, F\mathbf{g}_2, \ldots, F^{\nu_2 - 1}\mathbf{g}_2.$$

If $\nu_1 + \nu_2 = n$ then the procedure stops; otherwise consider $\mathbf{g}_3$ and so on, till one eventually gets an n-tuple of linearly independent vectors

$$\mathbf{g}_1, \ldots, F^{\nu_1 - 1}\mathbf{g}_1, \mathbf{g}_2, \ldots, F^{\nu_2 - 1}\mathbf{g}_2, \ldots \mathbf{g}_t, \ldots, F^{\nu_t - 1}\mathbf{g}_t,$$

with $\nu_1 + \nu_2 + \ldots + \nu_t = n$.

2. Introduce the invertible matrix $Q \in \mathbb{R}^{n\times n}$

$$Q = [\,\mathbf{g}_1 \quad \dots \quad F^{\nu_1-1}\mathbf{g}_1 \;\mid\; \mathbf{g}_2 \quad \dots \quad F^{\nu_2-1}\mathbf{g}_2 \;\mid\; \dots \quad F^{\nu_{t-1}-1}\mathbf{g}_{t-1} \;\mid\; \mathbf{g}_t \quad \dots \quad F^{\nu_t-1}\mathbf{g}_t\,] \tag{6.36}$$

and the matrix $S \in \mathbb{R}^{m\times n}$

$$S \;:=\; \Big[\; \mathbf{0} \;\; \dots \;\; \underset{\substack{\uparrow\\ \nu_1-\text{th}\\ \text{column}}}{\mathbf{e}_2} \;\mid\; \mathbf{0} \;\; \dots \;\; \underset{\substack{\uparrow\\ \nu_1+\nu_2-\text{th}\\ \text{colum}}}{\mathbf{e}_3} \;\mid\; \mathbf{0} \;\; \dots\dots \;\; \underset{\substack{\uparrow\\ \nu_1+\ldots+\nu_{t-1}-\text{th}\\ \text{column}}}{\mathbf{e}_t} \;\mid\; \mathbf{0} \;\; \dots \;\; \mathbf{0} \;\Big] \tag{6.37}$$

where $\mathbf{e}_i$ is the i-th column of the identity matrix $m \times m$.

3. Define the matrix $M \in \mathbb{R}^{m\times n}$ as

$$M := SQ^{-1}. \tag{6.38}$$

The matrix M built in the last step of the above procedure satisfies Heymann's lemma, i.e., it is such that $(F + GM, \mathbf{g}_1)$ is reachable. Indeed, from the definition of M one has

$$\begin{aligned} MQ &= M\,[\,\mathbf{g}_1 \;\dots\; F^{\nu_1-1}\mathbf{g}_1 \mid \mathbf{g}_2 \;\dots\; F^{\nu_2-1}\mathbf{g}_2 \mid \dots\; F^{\nu_{t-1}-1}\mathbf{g}_{t-1} \mid \mathbf{g}_t \;\dots\; F^{\nu_t-1}\mathbf{g}_t\,] \\ &= [\,\mathbf{0} \;\dots\; \mathbf{e}_2 \mid \mathbf{0} \;\dots\; \mathbf{e}_3 \mid \dots\; \mathbf{e}_t \mid \mathbf{0} \;\dots\; \mathbf{0}\,] \\ &= S. \end{aligned} \tag{6.39}$$

Moreover the reachability of $(F + GM, \mathbf{g}_1)$, i.e., the linear independence of the various $\mathbf{g}_1, (F + GM)\mathbf{g}_1, \dots, (F + GM)^{n-1}\mathbf{g}_1$ follows directly from (6.39). Indeed:

$$\begin{aligned} \mathbf{g}_1 &= \mathbf{g}_1 \\ (F+GM)\mathbf{g}_1 &= F\mathbf{g}_1 + GM\mathbf{g}_1 = F\mathbf{g}_1 \\ \dots\dots &= \dots\dots \\ (F+GM)^{\nu_1-1}\mathbf{g}_1 &= (F+GM)F^{\nu_1-2}\mathbf{g}_1 = F^{\nu_1-1}\mathbf{g}_1 + GMF^{\nu_1-2}\mathbf{g}_1 = F^{\nu_1-1}\mathbf{g}_1 \\ (F+GM)^{\nu_1}\mathbf{g}_1 &= (F+GM)F^{\nu_1-1}\mathbf{g}_1 = F^{\nu_1}\mathbf{g}_1 + GMF^{\nu_1-1}\mathbf{g}_1 \\ &= F^{\nu_1}\mathbf{g}_1 + G\mathbf{e}_2 = \mathbf{g}_2 + (*) \\ (F+GM)^{\nu_1+1}\mathbf{g}_1 &= (F+GM)(\mathbf{g}_2 + (*)) = F\mathbf{g}_2 + GM\mathbf{g}_2 + (F+GM)(*) \\ &= F\mathbf{g}_2 + (*) \\ \dots\dots &= \dots\dots \\ (F+GM)^{n-1}\mathbf{g}_1 &= (F+GM)(F^{\nu_t-2}\mathbf{g}_t + (*)) = F^{\nu_t-1}\mathbf{g}_t + (*) \end{aligned}$$

where the terms denoted by $(*)$ are linear combinations of the preceding vectors. ■

Once one has obtained the reachable pair $(F + GM, \mathbf{g}_i)$, it is possible to perform the second step of the whole procedure: compute a row matrix K_i that allocates the eigenvalues of the matrix $(F + GM) + \mathbf{g}_i K_i$ as desired by letting

$$\bar{K} := \begin{bmatrix} 0 \\ \vdots \\ K_i \\ \vdots \\ 0 \end{bmatrix}.$$

In this way, indeed, it is possible to allocate the eigenvalues of $(F + GM) + \mathbf{g}_i K_i = (F + GM) + G\bar{K} = F + G(M + \bar{K})$ wherever desired in $\mathbb{C}$.

The two feedback matrices M and $\bar{K}$ that we computed separately in the two steps above can be summed up to give a single feedback matrix $K = M + \bar{K}$. The block scheme in Figure 6.5.1 presents then the resulting scheme for a particular multi-input system with $m = 2$ and $n = 3$.

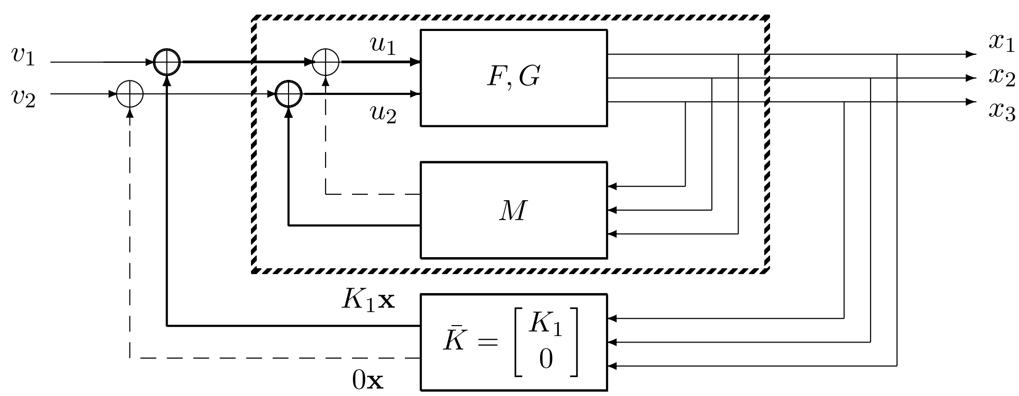


Figure 6.5.1

Example 6.5.1. The system Σ with

$$F = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

is reachable, but not from a single input. Consider the matrices Q and S:

$$Q = [\, \mathbf{g}_1 \quad F\mathbf{g}_1 \quad F^2\mathbf{g}_1 \quad \mathbf{g}_2 \,] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$S = [\, \mathbf{0} \quad \mathbf{0} \quad \mathbf{e}_2 \quad \mathbf{0} \,] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

From Q and S we can build the state feedback matrix

$$M = SQ^{-1} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

that makes the system

$$(F+GM, \mathbf{g}_1) = \left(\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

reachable. Let then $p(s) = s^4 + p_3 s^3 + p_2 s^2 + p_1 s + p_0$ be the desired characteristic polynomial for the to-be-designed closed loop system. Since the pair $(F+GM, \mathbf{g}_1)$ is in controllable canonical form, it is possible to compute directly

$$K_1 = [\, k_0 \quad k_1 \quad k_2 \quad k_3 \,]$$

by imposing that the characteristic polynomial of

$$(F+GM) + \mathbf{g}_1 K_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ k_0 & k_1 & k_2+1 & k_3 \end{bmatrix}$$

is equal to $p(s)$. Since the characteristic polynomial of $(F+GM) + \mathbf{g}_1 K_1$ is $\Delta_{(F+GM)+\mathbf{g}_1 K_1}(s) = s^4 - k_3 s^3 - (k_2+1)s^2 - k_1 s - k_0$, we obtain

$$K_1 = [\, -p_0 \quad -p_1 \quad -(1+p_2) \quad -p_3 \,].$$

Therefore the overall feedback matrix is

$$K = M + \begin{bmatrix} K_1 \\ 0 \end{bmatrix} = \begin{bmatrix} -p_0 & -p_1 & -(1+p_2) & -p_3 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

To conclude, the eigenvalues of non-reachable subsystems cannot be modified by means of a state feedback, as proved in Proposition 6.1.2. At the same time, the eigenvalues of the reachable subsystems can be arbitrarily allocated, due to Heymann's lemma and Proposition 6.4.1. Therefore we get the following corollary:

Proposition 6.5.2 [Allocability of the eigenvalues and reachability of the system] *A necessary and sufficient condition for a system (F, G, H) to be fully reachable is that the eigenvalues of $F+GK$ can be arbitrarily allocated by varying K.* ■

Exercise 6.5.1 [Controlling a continuous time system with polynomial inputs] Consider the continuous-time reachable system $\dot{\mathbf{x}} = F\mathbf{x} + G\mathbf{u}$ of dimension n. Choosing an arbitrary but not degenerate time interval $[0, t]$, an initial condition $\mathbf{x}(0) = \mathbf{x}_0$ and a final state $\mathbf{x}(t) = \mathbf{x}_f$, prove that there exists a polynomial input $\mathbf{u}(\sigma)$ (i.e., an input whose entries are polynomial functions of the variable σ with $\sigma \in [0, t]$) that brings the closed loop system from the initial condition $\mathbf{x}_0$ to the desired final state $\mathbf{x}_f$.

♯ *Solution: If the matrix F is nilpotent, then the elements of the exponential matrix $e^{F\sigma}$ are polynomials in σ with degree smaller than n. Thus the components of the minimal-energy input $\mathbf{u}(\sigma) = G^T e^{F^T(t-\sigma)} \mathcal{W}_t^{-1}(\mathbf{x}_f - e^{Ft}\mathbf{x}_0)$ that have been obtained in (6.62) are polynomials of order smaller than n in the variable σ. Polynomials in σ (of which degree?) are also the components of the state evolution, since*

$$\mathbf{x}(\sigma) = e^{F\sigma}\mathbf{x}_0 + \int_0^\sigma e^{F(\sigma-\xi)} G\mathbf{u}(\xi) d\xi, \ \ \sigma \in [0, t].$$

If F is not nilpotent, then due to Proposition 6.5.2 there exists a feedback matrix K such that $\bar{F} := F + GK$ is nilpotent. Since the system $(\bar{F}, G)$ is still reachable, there exists a polynomial input $\mathbf{v}(\sigma)$ that brings the closed loop system $\dot{\mathbf{x}} = (F+GK)\mathbf{x} + G\mathbf{v}$ from $\mathbf{x}(0) = \mathbf{x}_0$ to $\mathbf{x}(t) = \mathbf{x}_f$. The state evolution $\mathbf{x}(\sigma)$ is also polynomial, thus also the input $\mathbf{u}(\sigma) := \mathbf{v}(\sigma) + K\mathbf{x}(\sigma)$, that controls the open loop system from $\mathbf{x}_0$ to $\mathbf{x}_f$, is a polynomial in σ.

6.6 Control invariants and Rosenbrock's theorem

The reachability of (F, G) allows to arbitrarily allocate the eigenvalues of $F + GK$. This, however, does not mean that the matrix $F+GK$ of the closed loop system can be arbitrarily modified as K varies. For example, is it possible to attribute to $F+GK$ an arbitrary Jordan structure by suitably varying K? In this section we will determine which parameters can not be modified by varying K and to what extent the structure of $F+GK$ can be modified with respect to that of F.

The set of the Kronecker indexes is the most important set of invariants when considering state feedback strategies (and, by duality, also when considering state observer design or the realization of multi-input multi-output systems). To define these indexes we proceed as follows.

Consider a reachable system $\Sigma = (F, G, H)$ with reachability index r and define the integers $q, q_2, \ldots, q_r$ through the relations

$$\begin{aligned} q &=: \operatorname{rank}[\,G\,] \\ q + q_2 &=: \operatorname{rank}[\,G \quad FG\,] \\ q + q_2 + q_3 &=: \operatorname{rank}[\,G \quad FG \quad F^2G\,] \\ \cdots & \quad\;\; \cdots \\ q + q_2 + \ldots + q_r &=: \operatorname{rank}[\,G \quad FG \quad \ldots \quad F^{r-1}G\,] = n. \end{aligned} \tag{6.40}$$

By construction these integers satisfy the inequalities

$$q \geq q_2 \geq q_3 \geq \ldots \geq q_r. \tag{6.41}$$

Remark 1 The integer $q_i,\ i = 2, 3, \ldots$ represents the maximum number of columns in the block $F^{i-1}G$ that are linearly independent of the columns of $\mathcal{R}_{i-1}$. In other words, if $\mathcal{B}_{i-1}$ is a basis for the vector space $\mathrm{Im}\mathcal{R}_{i-1}$ then this basis can be completed to another basis $\mathcal{B}_i$ for the space $\mathrm{Im}\mathcal{R}_i$ by adding to it a q_i-tuple of the type

$$F^{i-1}\mathbf{g}_{\nu_1}, F^{i-1}\mathbf{g}_{\nu_2}, \ldots, F^{i-1}\mathbf{g}_{\nu_{q_i}}. \tag{6.42}$$

Obviously, every other column $F^{i-1}\mathbf{g}_k$ of $F^{i-1}G$ with $k \neq \nu_1, \nu_2, \ldots, \nu_{q_i}$ is a linear combination of the vectors in (6.42) and of $\mathcal{B}_{i-1}$.
Considering now $\mathcal{R}_{i+1}$, it is immediate to see that each of its columns is a linear combination of the vectors in $\mathcal{B}_i$ and of the vectors

$$F^{i}\mathbf{g}_{\nu_1}, F^{i}\mathbf{g}_{\nu_2}, \ldots, F^{i}\mathbf{g}_{\nu_{q_i}}, \tag{6.43}$$

since the columns $F^i\mathbf{g}_k,\ k \neq \nu_1, \nu_2, \ldots, \nu_{q_i}$ can be expressed combining the vectors of (6.43) and the vectors in $F\mathcal{B}_{i-1}$ (i.e., the vectors of (6.43) and those of $\mathcal{B}_i$). This implies that $F^i\mathbf{g}_{\nu_1}, F^i\mathbf{g}_{\nu_2}, \ldots, F^i\mathbf{g}_{\nu_{q_i}}$ and the columns of $\mathcal{B}_i$ constitute a set of generators $\mathcal{G}_{i+1}$ for the space generated by the columns of $\mathcal{R}_{i+1}$. Possibly by eliminating from $\mathcal{G}_{i+1}$ some of the columns that are included in the list (6.43), we obtain in this way a basis $\mathcal{B}_{i+1}$ for $\mathrm{Im}\mathcal{R}_{i+1}$. It follows that:

1. q_{i+1} cannot be greater than q_i;
2. starting from a basis $\mathcal{B}_i$ of $\mathrm{Im}\mathcal{R}_i$ formed by columns of the matrix $\mathcal{R}_i$, to obtain a basis $\mathcal{B}_{i+1}$ of $\mathrm{Im}\mathcal{R}_{i+1}$ it is sufficent to add to $\mathcal{B}_i$ the F-images of some of the columns of $F^{i-1}G$ used in $\mathcal{B}_i$.

Once we obtained the columns of the reachability matrix in r steps

$$\mathcal{R}_r = [\,\mathbf{g}_1 \ \ \mathbf{g}_2 \ \ \cdots \ \ \mathbf{g}_m \quad F\mathbf{g}_1 \ \ F\mathbf{g}_2 \ \ \cdots \ \ F\mathbf{g}_m \quad \cdots \quad F^{r-1}\mathbf{g}_1 \ \ F^{r-1}\mathbf{g}_2 \ \ \cdots \ \ F^{r-1}\mathbf{g}_m\,]$$

we can sort them as follows:

$$\begin{array}{ccccc} \mathbf{g}_1 & \mathbf{g}_2 & \mathbf{g}_3 & \dots & \mathbf{g}_m \\ F\mathbf{g}_1 & F\mathbf{g}_2 & F\mathbf{g}_3 & \dots & F\mathbf{g}_m \\ F^2\mathbf{g}_1 & F^2\mathbf{g}_2 & F^2\mathbf{g}_3 & \dots & F^2\mathbf{g}_m \\ \dots & \dots & \dots & \dots & \dots \\ F^{r-1}\mathbf{g}_1 & F^{r-1}\mathbf{g}_2 & F^{r-1}\mathbf{g}_3 & \dots & F^{r-1}\mathbf{g}_m \end{array} \tag{6.44}$$

To obtain the Kronecker indexes we can adopt the following procedure:

1. from the first row of the table choose q linearly independent vectors;

2. from the second row, choose q_2 vectors that:

   - belong to the same columns of the table where the first q linearly independent vectors have been chosen;

   - are such that the union of the vectors chosen in the first and second rows are a basis for $\mathrm{Im}\,[\,G \quad FG\,]$;

3. from the third row, choose q_3 vectors that:

   - belong to the same columns of the table where the second q_2 linearly independent vectors have been chosen;

   - are such that the union of the vectors chosen in the first, second and third rows are a basis for $\mathrm{Im}\,[\,G \quad FG \quad F^2G\,]$;

4. continue with choosing $q_4, q_5, \dots$ vectors similarly as above until one gets n linearly independent vectors;

5. consider the columns in the table above where the n vectors $F^i\mathbf{g}_j$ have been chosen: these vectors form q vertical chains, one for each value j of the set of the indexes corresponding to the columns of G that have been chosen at the first step;

6. evaluate the length of each of these q chains (i.e., how many vectors belong to the chain). The chain should contain $\kappa_1, \kappa_2, \dots, \kappa_q$ vectors, respectively, with $\kappa_1 + \kappa_2 + \dots + \kappa_q = n$ and with $\max\{\kappa_1, \kappa_2, \dots, \kappa_q\} = r$, i.e., the reachability index of the system.

For instance

$$
\begin{array}{ccccccccccccc}
\mathbf{g}_1 & | & \mathbf{g}_2 & | & \mathbf{g}_3 & | & \mathbf{g}_4 & \cdots & | & \mathbf{g}_s & | & \cdots & \mathbf{g}_m \\
F\mathbf{g}_1 & | & F\mathbf{g}_2 & | & F\mathbf{g}_3 & | & F\mathbf{g}_4 & \vdots & | & F\mathbf{g}_s & | & \vdots & F\mathbf{g}_m \\
\vdots & | & \vdots & | & F^2\mathbf{g}_3 & | & F^2\mathbf{g}_4 & \vdots & | & F^2\mathbf{g}_s & | & \vdots & F^2\mathbf{g}_m \\
\vdots & | & F^{\kappa_1-1}\mathbf{g}_2 & | & \vdots & | & \vdots & \vdots & | & F^3\mathbf{g}_s & | & \vdots & F^3\mathbf{g}_m \\
 & & ----- & | & & | & & & | & & | & & \\
\vdots & & \vdots & | & F^{\kappa_2-1}\mathbf{g}_3 & | & \vdots & \vdots & | & \vdots & | & \vdots & \vdots \\
 & & & & ----- & & & & | & & | & & \\
\vdots & & \vdots & & \vdots & & \vdots & \vdots & | & F^{\kappa_q-1}\mathbf{g}_s & | & \vdots & \vdots \\
 & & & & & & & & & ----- & & & \\
\vdots & & \vdots & & \vdots & & \vdots & \vdots & & \vdots & & \vdots & \vdots \\
F^{r-1}\mathbf{g}_1 & & F^{r-1}\mathbf{g}_2 & & F^{r-1}\mathbf{g}_3 & & F^{r-1}\mathbf{g}_4 & \vdots & & F^{r-1}\mathbf{g}_s & & \vdots & F^{r-1}\mathbf{g}_m \\
 & & 1^{st}\text{chain} & & 2^{nd}\text{chain} & & & & & q\text{−thchain} & & &
\end{array}
$$

The integers $\kappa_1, \kappa_2, \dots, \kappa_q$ are called *Kronecker indexes* or *control invariants* of the reachable pair (F, G). Notice that q, the number of vectors selected in the first row of (6.44), and thus the number of vertical chains built in the algorithm above, coincides with the dimension q of $\mathrm{Im}[G]$. Similarly, q_2, the number of vectors selected in the second row of (6.44) and thus the number of vertical chains that contain at least two vectors, coincides with the dimension of $\mathrm{Im}\,[\,G \quad FG\,]$ minus the dimension of $\mathrm{Im}[G]$. Iterating, q_3 is simultaneously the number of vectors selected in the third row of (6.44), the number of chains that contain at least three vectors, and the dimension of $\mathrm{Im}\,[\,G \quad FG \quad F^2G\,]$ minus the dimension of $\mathrm{Im}\,[\,G \quad FG\,]$. For $q_4, q_5, \dots$ the concepts are the same.
As for the lengths of the various chains, the number of chains of length 1 is obviously $\ell_1 := q - q_2$, the number of chains of length 2 is $\ell_2 := q_2 - q_3$, and so on.

We thus constructively proved that there exist $\ell_i := q_i - q_{i+1}$ chains of length $i < r$ and $\ell_r = q_r$ chains of length r. Thus the family of the Kronecker indexes κ_i is given by

$$\{\kappa_1, \kappa_2, \dots, \kappa_q\} = \{\, \underbrace{1, \dots, 1}_{\ell_1 \text{ times}}, \underbrace{2, \dots, 2}_{\ell_2 \text{ times}}, \dots, \underbrace{r, \dots, r}_{\ell_r \text{ times}} \,\}$$

We can thus conclude that the r-tuple of sorted integers $(q_1, q_2, \dots, q_r)$ determines and is univoquely determined by the r-tuple of sorted integers $(\ell_1, \ell_2, \dots, \ell_r)$. At the same time, $(\ell_1, \ell_2, \dots, \ell_r)$ is determined by the q-tuple of non-sorted indexes $\{\kappa_1, \kappa_2, \dots, \kappa_q\}$. This means that the Kronecker indexes do not depend on the specific vectors chosen in the various rows of (6.44).

Proposition 6.6.1 [Invariance of the Kronecker indexes] *The Kronecker indexes κ_i of a generic reachable system (F, G) are invariant with respect to both the algebraic equivalence of systems and state feedback operations. I.e., $(T^{-1}(F + GK)T, T^{-1}G)$ and (F, G) share the same Kronecker indexes for every nonsingular transformation matrix T and feedback matrix K.*
Proof The invariance with respect to the algebraic equivalence is immediate. As far as the invariance with respect to state feedback operations is concerned, we now prove by

induction that for every i one has

$$\mathrm{Im}\,[\,G \quad FG \quad \ldots \quad F^{i-1}G\,] = \mathrm{Im}\,[\,G \quad (F+GK)G \quad \ldots \quad (F+GK)^{i-1}G\,]\,. \tag{6.45}$$

(6.45) is obvious for $i=1$. If (6.45) holds for $i=h$, this means that

$$\begin{aligned}
& \mathrm{Im}\,[\,G \quad (F+GK)G \quad \ldots \quad (F+GK)^{h}G\,] \\
= \; & \mathrm{Im}\,[\,G \quad (F+GK)G \quad \ldots \quad (F+GK)^{h-1}G\,] + \mathrm{Im}\,[\,(F+GK)^{h}G\,] \\
= \; & \mathrm{Im}\,[\,G \quad FG \quad \ldots \quad F^{h-1}G\,] + (F+GK)\mathrm{Im}\,[\,(F+GK)^{h-1}G\,] \\
\subseteq \; & \mathrm{Im}\,[\,G \quad FG \quad \ldots \quad F^{h-1}G\,] + (F+GK)\mathrm{Im}\,[\,G \quad FG \quad \ldots \quad F^{h-1}G\,] \\
\subseteq \; & \mathrm{Im}\,[\,G \quad FG \quad \ldots \quad F^{h-1}G\,] + \mathrm{Im}\,[\,FG \quad F^2G \quad \ldots \quad F^{h}G\,] + \mathrm{Im}\,[\,G\,] \\
= \; & \mathrm{Im}\,[\,G \quad FG \quad \ldots \quad F^{h-1}G \quad F^{h}G\,]\,.
\end{aligned}$$

Notice that the reverse inclusion can be obtained by assuming $F = (F+GK) + G(-K)$. Thus (6.45) holds also for $i = h+1$ and thus, by induction, for every i. ■

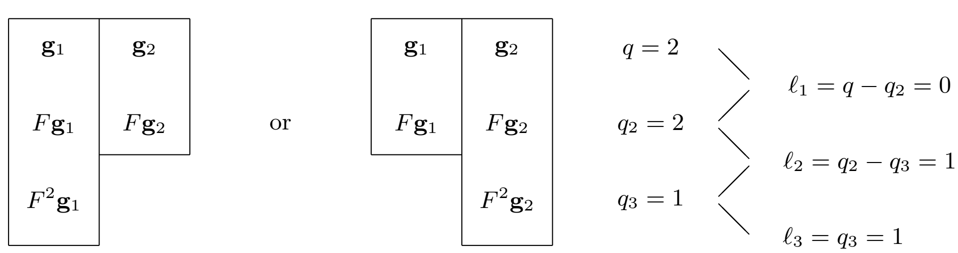
Example 6.6.1 Consider the pair (F,G) with

$$F = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad G = \begin{bmatrix} 2 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}.$$

Its reachability matrix is

$$\begin{array}{c} \mathcal{R} = \left[\begin{array}{cccccccc} 2 & 0 & 0 & 0 & 1 & 0 & & \\ 0 & 0 & 1 & 0 & 0 & 0 & & \\ 1 & 0 & 0 & 0 & 1 & 1 & \cdots & \\ 0 & 0 & 1 & 1 & 0 & 0 & & \\ 1 & 1 & 0 & 0 & 0 & 0 & & \end{array} \right] \\ \begin{array}{cccccc} \quad \mathbf{g}_1 & \mathbf{g}_2 & F\mathbf{g}_1 & F\mathbf{g}_2 & F^2\mathbf{g}_1 & F^2\mathbf{g}_2 \end{array} \end{array}$$

so that there are two possible diagrams, as shown below, from which we can infer that the Kronecker indexes are $\kappa_1 = 3$ and $\kappa_2 = 2$.



If the pair (F,G) is reachable, as we proved before, the characteristic polynomial of $F+GK$ can be arbitrarily allocated. At the same time, the Kronecker indexes cannot be modified through K. Thus, can the matrix $F+GK$ be made similar to an arbitrary matrix? In other words, can we choose K so that $F+GK$ represents an arbitrary linear transformation with respect to an appropriate basis for the state space?

Note that the fact that the characteristic polynomial $\Delta_{F+GK}(s)$ can be arbitrarily allocated does not imply that $F+GK$ can be an arbitrary linear transformation, since there exist matrices that share the same characteristic polynomial but are not similar (or, in a more abstract terms, since the characteristic polynomial is not a complete invariant for the equivalence relation induced by the operation of algebraic equivalence between systems). The following example captures exactly this fact through some considerations on a single-input system:

Example 6.6.2 The pair $(F, \mathbf{g})$ defined by

$$F = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

is fully reachable, so that the characteristic polynomial of $F + gK$ can be arbitrarily allocated. In particular, assuming $K = \bar{K} = [-1 \quad 2]$ we impose the characteristic polynomial to be equal to $(s-1)^2$.

Now there are two different Jordan forms that admit $(s-1)^2$ as a characteristic polynomial:

$$J_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad J_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Since $(F + \mathbf{g}K, \mathbf{g})$ is reachable, $F + \mathbf{g}K$ is cyclic for every choice of K. Thus $F + \mathbf{g}K$ cannot be similar to J_2. Notice also that for $K = \bar{K}$ the matrix $F + \mathbf{g}K$ is similar to J_1.

Since two matrices are similar if and only if they share all their invariant polynomials, to understand to which similarity class the matrix $F + GK$ belongs to it is sufficient to investigate which invariant polynomials of $F + GK$ can exhibit, as K varies.

The solution to this problem for the case of single-input systems is particularly simple. Indeed the reachability of $(F, \mathbf{g})$ implies the reachability of $(F + \mathbf{g}K, \mathbf{g})$ (and thus the fact that $F + \mathbf{g}K$ is cyclic). The matrix $F + \mathbf{g}K$ has thus only one invariant polynomial that coincides with both the minimal and the characteristic polynomials and that can be arbitrarily allocated due to Proposition 6.4.1. To summarise:

Proposition 6.6.2 *Let $(F, \mathbf{g})$ be a n-dimensional single-input reachable. Then for every $K \in \mathbb{R}^{1\times n}$ the matrix $(F + \mathbf{g}K)$ has only one invariant polynomial of degree n. At the same time for every monic polynomial $p(s)$ of degree n it is possible to find an appropriate $K \in \mathbb{R}^{1\times n}$ such that the invariant polynomial of $F + \mathbf{g}K$ is $p(s)$.* ■

For multi-input systems the solution is instead given by the following theorem, based on the Kronecker indexes.

Proposition 6.6.3 [Rosenbrock's theorem] *Let $\Sigma = (F, G, -)$ a reachable system of dimension n and let $\kappa_1 \geq \kappa_2 \geq \ldots \geq \kappa_q$ be its Kronecker indexes. Then for a generic choice of the feedback matrix $K \in \mathbb{R}^{m\times n}$ the cyclic index c and the invariant polynomials $\psi_1(s), \psi_2(s), \ldots, \psi_c(s)$ (with $\psi_i(s)|\psi_{i-1}(s)$) of $F + GK$ satisfy the conditions*

$$c \leq q$$

$$\begin{aligned} \deg\, \psi_1 &\geq \kappa_1 \\ \deg\, \psi_1 + \deg\, \psi_2 &\geq \kappa_1 + \kappa_2 \\ \deg\, \psi_1 + \deg\, \psi_2 + \deg\, \psi_3 &\geq \kappa_1 + \kappa_2 + \kappa_3 \\ \vdots \quad &\geq \quad \vdots \end{aligned} \tag{6.46}$$

At the same time, if the index $c \leq q$ and the monic polynomials $p_1(s), p_2(s), \ldots, p_c(s)$ satisfying

$$\begin{array}{rcl} p_i(s)|p_{i-1}(s) & , & 1 < i \leq c \\ \sum_{i=1}^{c} \deg\ p_i & = & n \\ \deg\ p_1 & \geq & \kappa_1 \\ \deg\ p_1 + \deg\ p_2 & \geq & \kappa_1 + \kappa_2 \\ \deg\ p_1 + \deg\ p_2 + \deg\ p_3 & \geq & \kappa_1 + \kappa_2 + \kappa_3 \\ \vdots & \geq & \vdots \end{array} \tag{6.47}$$

are arbitrarily fixed, then there exists a feedback matrix K such that the invariant polynomials of $F + GK$ are $p_1(s), p_2(s), \ldots, p_c(s)$. ■

We omit proving the theorem, but comment on its consequences. The constraints posed by Rosenbrock's theorem on the invariant polynomials can be interpreted as constraints on the Jordan form of $F + GK$. More precisely:

1. condition $c \leq q$ follows from the reachability of $(F+GK, G)$ and from Corollary 6.7.2. This condition implies that it is not possible to have more than q mini-blocks relative to the same eigenvalue;

2. deg $\psi_1 \geq \kappa_1$ implies that the sum of the dimensions of the mini-blocks of maximal dimension relative to each eigenvalue (and thus the degree of the minimal polynomial) cannot be smaller than κ_1;

3. deg ψ_1 + deg $\psi_2 \geq \kappa_1 + \kappa_2$ implies that the sum of the dimensions of the mini-blocks of maximal dimension plus the sum of the dimensions of the mini-blocks with immediately smaller dimension relative to each eigenvalue cannot be smaller than $\kappa_1 + \kappa_2$;

4. deg ψ_1 + deg ψ_2 + deg $\psi_3 \geq \kappa_1 + \kappa_2 + \kappa_3$ and the subsequent conditions have similar interpretations as above.

Example 6.6.3 Consider a reachable continuous-time system $\Sigma = (F, G, H)$ with dimension n and with $H = I_n$. We want to determine under what conditions there exists a state feedback matrix $K \in \mathbb{R}^{m \times n}$ for which, independently of the initial state, the unforced output evolution is constant in time.

Since the outputs coincide with the states, the assignment requires $\dot{\mathbf{x}}(t) = (F + GK)\mathbf{x}(t)$ to have solutions that are constant in time for every initial condition, i.e., K should make $F + GK$ be the null matrix.

Since the cyclic index of the null matrix is n, then for the Rosenbrock theorem it must be $q = n$. I.e., G must have rank n. Viceversa, it is obvious that if G has rank n then by varying K we can make $F + GK$ equal to any element of $\mathbb{R}^{n \times n}$, and thus also the null matrix.

Example 6.6.4 Consider the reachable discrete-time pair (F, G) with

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

One wants to determine, if possible, a state feedback matrix K such that starting from a generic non-null initial state the unforced state evolution of the closed loop system becomes zero in one step. This condition corresponds to choose K so that $F + GK$ is the null matrix. Since the cyclic index of the matrix 0_3 is 3 and is thus greater than the rank $q = 2$ of the matrix G, this problem cannot be solved. Alternatively, we can observe that the Kronecker indexes of (F, G) are $\kappa_1 = 2,\ \kappa_2 = 1$. Allocating the minimal polynomial of $F + GK$ so that it becomes z (whose degree is 1) is thus not possible. Verify that the problem can be solved if one wants to make the state go to zero in two steps instead of in just one.

6.7 Canonical forms and allocation of the eigenvalues: case $m > 1$

The crucial point in the procedure shown in Section 6.5 about how to allocate the eigenvalues of the closed loop system is based on Heymann's lemma. As we saw before, this procedure consists of obtaining, through an opportune feedback matrix, a system that is reachable from just one input. This means that the matrix $F + GK$ that solves the eigenvalues allocation problem is always cyclic and thus endowed of a single invariant polynomial. Rosenbrock's theorem, on the other hand, guarantees that for multivariable systems the class of matrices $F + GK$ that can be obtained when varying K is generally richer than the class of cyclic matrices. Thus the procedure based on Heymann's lemma, even if solving the allocation problem, returns only a subset of all possible solutions.
In this section we thus introduce another technique to allocate the eigenvalues that is based on the introduction of a suitable controllable canonical form for multi-input systems. This technique allows, in particular, the determination of a feedback matrix K that allocates all the invariant polynomials of $F + GK$ so that they coincide with any set of polynomials $p_i(s)$ that simultaneously satisfy $p_i(s) \mid p_{i-1}(s)$ and the limits imposed by Rosenbrock's theorem, i.e.,

$$\deg\, p_1 = \kappa_1, \quad \deg\, p_1 + \deg\, p_2 = \kappa_1 + \kappa_2, \quad \deg\, p_1 + \deg\, p_2 + \deg\, p_3 = \kappa_1 + \kappa_2 + \kappa_3, \ldots$$

In this way we can obtain a matrix $F + GK$ whose cyclic index is maximal and whose minimal polynomial has the smallest possible degree.

This procedure exploits the *multivariable controllable canonical form* that can be obtained by properly transforming a reachable system (F, G) through the following transformation matrix T: assume, for the sake of simplicity, that the maximal Kronecker index κ_1 is equal to the length of the chain associated to $\mathbf{g}_1$, that κ_2 is equal to the length of the chain associated to $\mathbf{g}_2$, and so on. Consider then the nonsingular matrix

$$P = [\,\mathbf{g}_1 \quad \ldots \quad F^{\kappa_1 - 1}\mathbf{g}_1 \quad \mathbf{g}_2 \quad \ldots \quad F^{\kappa_2 - 1}\mathbf{g}_2 \quad \ldots\ldots \quad \mathbf{g}_q \quad \ldots \quad F^{\kappa_q - 1}\mathbf{g}_q\,]\,.$$

Let $\mathbf{c}_1^T, \mathbf{c}_2^T, \ldots, \mathbf{c}_q^T$ the rows of the matrix P^{-1} corresponding to the κ_1-th, $\kappa_1 + \kappa_2$-th, ...,

$\kappa_1 + \kappa_2 + \ldots + \kappa_q = n$-th indexes. With these rows build then the matrix

$$
T^{-1} = \begin{array}{c}
\left[\begin{array}{c}
\mathbf{c}_1^T \\ \vdots \\ \mathbf{c}_1^T F^{\kappa_1 - 1} \\
\mathbf{c}_2^T \\ \vdots \\ \mathbf{c}_2^T F^{\kappa_2 - 1} \\
\vdots \\
\mathbf{c}_q^T \\ \vdots \\ \mathbf{c}_q^T F^{\kappa_q - 1}
\end{array}\right]
\end{array}
\begin{array}{l}
\left.\begin{array}{c} \\ \\ \\ \end{array}\right\} \kappa_1 \text{ rows} \\
\left.\begin{array}{c} \\ \\ \\ \end{array}\right\} \kappa_2 \text{ rows} \\
\quad \vdots \\
\left.\begin{array}{c} \\ \\ \\ \end{array}\right\} \kappa_q \text{ rows}
\end{array}
$$

By resorting to this transformation matrix we thus obtain the following multivariable controllable canonical form $(F_c, G_c) = (T^{-1} F T, T^{-1} G)$, where the entries denoted by $\star$ denote generic values:

$$
F_c = \left[\begin{array}{cccc|cccc|ccc|cccc}
0 & 1 & & & & & & & & & & & & & \\
 & & \ddots & & & & & & & & & & & & \\
 & & & 1 & & & & & & & & & & & \\
\star & \star & \cdots & \star & \star & \star & \cdots & \star & \cdots & \cdots & \cdots & \star & \star & \cdots & \star \\
 & & \underbrace{\qquad}_{\kappa_1 \times \kappa_1} & & & & & & & & & & & & \\
 & & & & 0 & 1 & & & & & & & & & \\
 & & & & & & \ddots & & & & & & & & \\
 & & & & & & & 1 & & & & & & & \\
\star & \star & \cdots & \star & \star & \star & \cdots & \star & \cdots & \cdots & \cdots & \star & \star & \cdots & \star \\
 & & & & & & \underbrace{\qquad}_{\kappa_2 \times \kappa_2} & & & & & & & & \\
 & & & & & & & & \ddots & & & & & & \\
 & & & & & & & & & \ddots & & & & & \\
 & & & & & & & & & & \ddots & & & & \\
 & & & & & & & & & & & 0 & 1 & & \\
 & & & & & & & & & & & & & \ddots & \\
 & & & & & & & & & & & & & & 1 \\
\star & \star & \cdots & \star & \star & \star & \cdots & \star & \cdots & \cdots & \cdots & \star & \underbrace{\star}_{\kappa_q \times \kappa_q} & \cdots & \star
\end{array}\right]
$$

$$
G_c = \underbrace{\left[\begin{array}{c|ccccc|c}
0 & 0 & & & & 0 & \star \; \dots \; \star \\
\vdots & \vdots & & & & \vdots & \\
0 & 0 & & & & 0 & \\
1 & \star & & \star & \dots & \star & \star \; \dots \; \star \\
\hline
0 & 0 & & & & & \star \; \dots \; \star \\
\vdots & \vdots & & & & & \\
0 & 0 & & & & 0 & \\
0 & 1 & & \star & \dots & \star & \star \; \dots \; \star \\
\hline
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \; \cdot \; \cdot \\
\hline
0 & 0 & & & & 0 & \star \; \dots \; \star \\
\vdots & \vdots & & & & & \\
0 & 0 & & & & 0 & \\
0 & 0 & & \dots & & 1 & \star \; \dots \; \star
\end{array}\right]}_{q \text{ columns} \qquad\qquad m-q \text{ columns}}
\begin{array}{l}
\left.\vphantom{\begin{array}{c}0\\0\\0\end{array}}\right\}\kappa_1 \text{ rows} \\
\\
\left.\vphantom{\begin{array}{c}0\\0\\0\end{array}}\right\}\kappa_2 \text{ rows} \\
\\
\left.\vphantom{\begin{array}{c}0\\0\\0\end{array}}\right\}\kappa_q \text{ rows}
\end{array}
\tag{6.48}
$$

Removing in G_c all the rows except the ones with indexes $\kappa_1, \kappa_1+\kappa_2, \dots, \sum_{i=1}^{q} \kappa_i = n$, we get the $q \times m$ matrix

$$
\left[\begin{array}{ccccc|ccc}
1 & \alpha_{12} & \alpha_{13} & \dots & \alpha_{1q} & \star & \dots & \star \\
0 & 1 & \alpha_{23} & \dots & \alpha_{2q} & \star & \dots & \star \\
 & & & \ddots & \vdots & & \dots & \\
0 & 0 & 0 & \dots & 1 & \star & \dots & \star
\end{array}\right]
$$

One can then extract the first q columns of this matrix so to build the invertible $m \times m$ matrix

$$
N := \left[\begin{array}{ccccc|c}
1 & \alpha_{12} & \alpha_{13} & \dots & \alpha_{1q} & \\
0 & 1 & \alpha_{23} & \dots & \alpha_{2q} & \\
 & & & & & 0 \\
 & & & \ddots & \vdots & \\
0 & 0 & 0 & \dots & 1 & \\
\hline
 & & 0 & & & I_{m-q}
\end{array}\right]^{-1} . \tag{6.49}
$$

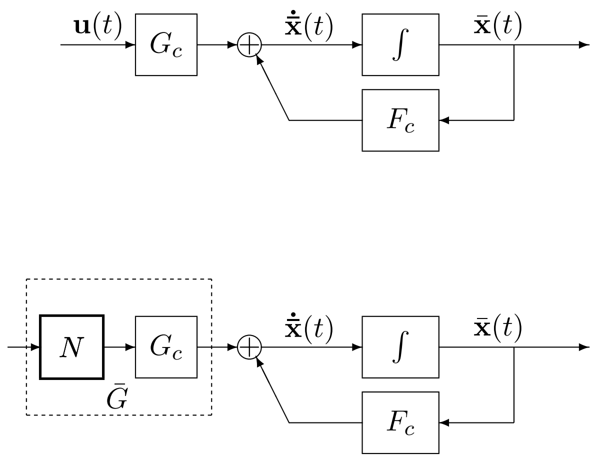


Figure 6.7.1

If the matrix N is used as a static filter to be applied to the m inputs of the system, then the resulting system becomes $(F_c, \bar{G}) = (F_c, G_cN)$, that differs from (F_c, G_c) only because of the input matrix (see Figure 6.7.1).

$$\bar{G} \;=\; G_cN = \left[\begin{array}{cccccc||ccc}
0 & 0 & & & & 0 & \star & \dots & \star \\
\vdots & \vdots & & & & \vdots & & & \\
0 & 0 & & & & 0 & & & \\
1 & 0 & 0 & \dots & & 0 & \star & \dots & \star \\
\hline
0 & 0 & & & & & \star & \dots & \star \\
\vdots & \vdots & & & & & & & \\
0 & 0 & & & & 0 & & & \\
0 & 1 & 0 & \dots & & 0 & \star & \dots & \star \\
\hline
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\hline
0 & 0 & & & & 0 & \star & \dots & \star \\
\vdots & \vdots & & & & & & & \\
0 & 0 & & & & 0 & & & \\
0 & 0 & \dots & & & 1 & \star & \dots & \star
\end{array}\right]
\begin{array}{l}
\left.\vphantom{\begin{array}{c}0\\0\\0\\0\end{array}}\right\}\kappa_1 \text{ rows} \\
\left.\vphantom{\begin{array}{c}0\\0\\0\\0\end{array}}\right\}\kappa_2 \text{ rows} \\
\\
\left.\vphantom{\begin{array}{c}0\\0\\0\\0\end{array}}\right\}\kappa_q \text{ rows}
\end{array}
\tag{6.50}$$

$$\underbrace{\qquad\qquad\qquad\qquad}_{q \text{ columns}} \qquad \underbrace{\qquad\qquad}_{m-q \text{ columns}}$$

Consider then the pair $(F_c, \bar{G})$. Our aim is to apply a feedback matrix whose last

$m - q$ rows are null. It is then immediate to verify that the matrix $\bar{G}\bar{K} \in \mathbb{R}^{n\times n}$ is null everywhere except for the rows of indexes $\kappa_1, \kappa_1 + \kappa_2, \dots, n$, and that those rows can have arbitrary values as $\bar{K}$ varies:

$$\bar{K} = \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1n} \\ k_{21} & k_{22} & \dots & k_{2n} \\ & \dots & \dots & \\ k_{q1} & k_{q2} & \dots & k_{qn} \\ - & - & - & - \\ & & 0 & \\ & & & \end{bmatrix} \implies \bar{G}\bar{K} = G_c(N\bar{K}) = \begin{bmatrix} 0 & 0 & & & 0 \\ \vdots & \vdots & & & \vdots \\ 0 & 0 & & & 0 \\ k_{11} & k_{12} & k_{13} & \dots & k_{1n} \\ 0 & 0 & & & 0 \\ \vdots & \vdots & & & \vdots \\ 0 & 0 & & & 0 \\ k_{21} & k_{22} & k_{23} & \dots & k_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & & & 0 \\ \vdots & \vdots & & & \\ 0 & 0 & & & 0 \\ k_{q1} & k_{q2} & k_{q3} & \dots & k_{qn} \end{bmatrix} \begin{array}{l} \\ \}\kappa_1 \text{ rows} \\ \\ \\ \\ \}\kappa_2 \text{ rows} \\ \\ \\ \vdots \\ \\ \\ \}\kappa_q \text{ rows} \\ \\ \end{array}$$

As soon as one appropriately selects the elements k_{ij}, the matrix $F_c + \bar{G}\bar{K}$ can thus be made block diagonal with diagonal blocks being companion matrices of dimension $\kappa_1 \times \kappa_1$, $\kappa_2 \times \kappa_2$, and so on, and each one of these blocks has arbitrary characteristic polynomial.

$$F_c + \bar{G}\bar{K} = \left[\begin{array}{cccc|cccc|c|cccc} 0 & 1 & & & & & & & & & & & \\ & & \ddots & & & & & & & & & & \\ & & & 1 & & & & & & & & & \\ \star & \star & \cdots & \star & & & & & & & & & \\ \hline & & & & 0 & 1 & & & & & & & \\ & & & & & & \ddots & & & & & & \\ & & & & & & & 1 & & & & & \\ & & & & \star & \star & \cdots & \star & & & & & \\ \hline & & & & & & & & \ddots & & & & \\ & & & & & & & & \ddots & & & & \\ \hline & & & & & & & & & 0 & 1 & & \\ & & & & & & & & & & & \ddots & \\ & & & & & & & & & & & & 1 \\ & & & & & & & & & \star & \star & \dots & \star \end{array}\right] \qquad (6.51)$$

In particular, the procedure allows to obtain a matrix $F_c + \bar{G}\bar{K}$ that has an arbitrary q-tuple of invariant polynomials $p_1(s),\ p_2(s), \dots p_q(s)$ with degrees respectively $\kappa_1,\ \kappa_2, \dots, \kappa_q$. In other words, the procedure allows to satisfy the conditions (6.47) in Rosenbrock's theorem as equalities.

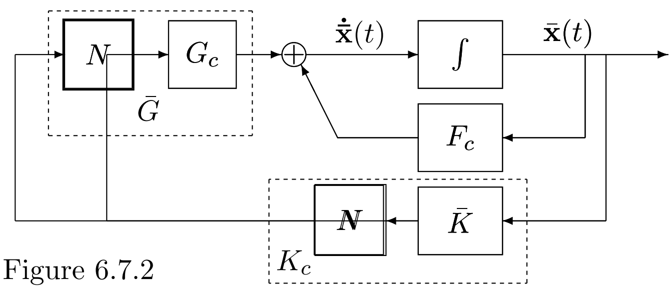


Figure 6.7.2

We finally notice that if we define

$$K_c = N\bar{K}$$

then K_c represents the feedback matrix to be applied to the open loop system expressed in its multivariable controllable canonical form to obtain the desired invariant polynomials.

EXERCISE 6.7.1 Prove that every system (F_c, G_c) expressed in multivariable controllable canonical form (6.48) is reachable.

♯ *Hint: construct the reachability matrix using the first q columns of G_c.*

6.8 Stabilization through state feedback

One of the most important objectives that can be achieved using state feedback techniques is that of making a system asymptotically stable. Obviously if the couple (F, G) is reachable, then the problem of stabilizing the system can be solved with an arbitrary stability margin, since the eigenvalues of $F + GK$ can be arbitrarily allocated.

If (F, G) is not reachable, though, the problem of stabilizing a system can be solved only when the non-reachable subsystem is asymptotically stable. Indeed, while the eigenvalues of the reachable subsystem can be allocated arbitrarily, the ones of the non-reachable subsystem are invariant with respect to the introduction of state feedback matrices. This implies that, in this case, even if one may stabilise the entire system, the stability margin cannot be increased arbitrarily.

Example 6.8.1 Consider the pair $(F, \mathbf{g})$ with

$$F = \begin{bmatrix} -1 & -2 & 3 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Its reachability matrix,

$$\mathcal{R} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix},$$

has rank one, so that only one of the eigenvalues of the system can be arbitrarily allocated through an opportune state feedback matrix. Transforming the system using

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

we obtain the standard reachability form

$$\bar{F} = T^{-1}FT = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix} = \begin{bmatrix} 0 & | & -2 & 3 \\ - & - & - & - \\ 0 & | & 3 & -4 \\ 0 & | & 1 & -1 \end{bmatrix}, \qquad \bar{\mathbf{g}} = T^{-1}\mathbf{g} = \begin{bmatrix} \bar{\mathbf{g}}_1 \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} 1 \\ -- \\ 0 \\ 0 \end{bmatrix}.$$

From this we notice that the characteristic polynomial of the non-reachable subsystem is

$$\det(sI_2 - \bar{F}_{22}) = \det \begin{bmatrix} s-3 & 4 \\ -1 & s+1 \end{bmatrix} = (s-1)^2$$

so that the unique eigenvalue of $\bar{F}_{22}$ is $+1$, with algebraic multiplicity 2. Thus the system cannot be stabilized (neither if it is continuous-time, nor if it is discrete-time).

What has been said before holds both for discrete- and continuous-time systems. Discrete-time systems, though, have a peculiar property that has no continuous-time counterpart: if $\Sigma = (F, G, H)$ has dimension n and a feedback matrix K exists such that $F+GK$ has all the eigenvalues that are null, then the resulting system has *finite memory*, meaning that the state at time $t = 0$ can influence only the states at times $t \leq n-1$. This is a consequence of Cayley-Hamilton theorem, that implies $(F + GK)^n = 0$. In this situation, a disturbance acting on the system at time $t = 0$ will not propagate for more than n steps. Notice moreover that this may be a pessimistic bound, since n is not necessarily the smallest value of h for which $(F + GK)^h = 0$. In particular, when the pair (F, G) is reachable, it is possible to choose K so that the nilpotency index h coincides with the maximum of the Kronecker indexes.

Definition 6.8.1 [Dead-beat controller] *Let $\Sigma = (F, G, H)$ be a discrete-time system of dimension n. Then a matrix $K \in \mathbb{R}^{m\times n}$ that allocates all the eigenvalues of $F + GK$ in zero is called a dead-beat controller (d.b.c.) for Σ.*

The existence of a d.b.c. is a necessary but not sufficient condition for the reachability of a system, but it is a necessary and sufficient condition for the controllability of a system, that, for discrete-time systems, is a weaker property.

Proposition 6.8.2 [Existence of a d.b.c.] *Let $\Sigma = (F, G, H)$ be a discrete-time system. Then the following conditions are equivalent:*

i) *Σ is controllable;*

ii) *in the standard reachability form of Σ, all the eigenvalues of the non-reachable subsystem are zero;*

iii) *Σ admits a d.b.c.;*

iv) the matrix $[\, zI_n - F \quad G\,]$ *is of full rank for each* $z \in \mathbb{C} \setminus \{0\}$.

Proof (i) $\Rightarrow$ (ii): If the system Σ is controllable, then

$$\mathrm{Im}\,[\, G \quad FG \quad \dots \quad F^{n-1}G\,] \supseteq \mathrm{Im} F^n. \tag{6.52}$$

Changing the basis of the state space we can write the system in standard reachability form

$$\bar{F} = T^{-1}FT = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix}, \quad \bar{G} = T^{-1}G = \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix}, \tag{6.53}$$

where the pair $(\bar{F}_{11}, \bar{G}_1)$ is reachable. Relation (6.52) then translates as

$$\mathrm{Im} \begin{bmatrix} \bar{G}_1 & \bar{F}_{11}\bar{G}_1 & \cdots & \bar{F}_{11}^{n-1}\bar{G}_1 \\ 0 & 0 & \cdots & 0 \end{bmatrix} \supseteq \mathrm{Im} \begin{bmatrix} \bar{F}_{11}^n & \star \\ 0 & \bar{F}_{22}^n \end{bmatrix}. \tag{6.54}$$

Since the last entries of every vector in the left subspace are zero, the same must hold for the vectors in the right subspace. Thus $\bar{F}_{22}^n = 0$, and this implies that $\bar{F}_{22}$ is nilpotent. In other words, this implies that all the eigenvalues of the non-reachable subsystem are zero.

(ii) $\Rightarrow$ (iii): Since $(\bar{F}_{11}, \bar{G}_1)$ is reachable, there exists a matrix $\bar{K}_1$ for which the eigenvalues of $\bar{F}_{11} + \bar{G}_1\bar{K}_1$ are zero. Thus also all the eigenvalues of

$$\begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix} + \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix} [\, \bar{K}_1 \quad \star\,] = \begin{bmatrix} \bar{F}_{11} + \bar{G}_1\bar{K}_1 & \star \\ 0 & \bar{F}_{22} \end{bmatrix} \tag{6.55}$$

are zero, independently of the values of the elements denoted by $\star$ in $\bar{K} = [\, \bar{K}_1 \quad \star\,]$. This matrix is thus a d.b.c. for the system in its standard reachability form and

$$K = [\, \bar{K}_1 \quad \star\,]\, T^{-1}$$

is a d.b.c. for the system $\Sigma = (F, G, H)$.

(iii) $\Rightarrow$ (i): If K is a d.b.c. for Σ, then $(F + GK)^n = 0$. Thus, independently of the initial state of the system

$$\mathbf{x}(t+1) = F\mathbf{x}(t) + G\mathbf{u}(t), \tag{6.56}$$

the input $\mathbf{u}(t) = K\mathbf{x}(t),\ t = 0, 1, \dots, n-1$, given by

$$\begin{array}{rcl} \mathbf{u}(0) & = & K\mathbf{x}(0) \\ \mathbf{u}(1) & = & K\mathbf{x}(1) = K(F + GK)\mathbf{x}(0) \\ \mathbf{u}(2) & = & K\mathbf{x}(2) = K(F + GK)^2\mathbf{x}(0) \\ \dots & & \dots \\ \mathbf{u}(n-1) & = & K\mathbf{x}(n-1) = K(F + GK)^{n-1}\mathbf{x}(0) \end{array}$$

controls the system (6.56) at time instant n to

$$\mathbf{x}(n) = F\mathbf{x}(n-1) + G\mathbf{u}(n-1) = (F + GK)^n\mathbf{x}(0) = \mathbf{0}.$$

(ii) $\Leftrightarrow$ (iv): By Theorem 6.7.1, the rank of the PBH matrix is smaller than n for all and only the eigenvalues associated to the non-reachable subsystem, i.e., for all and only the eigenvalues of $\bar{F}_{22}$. It is thus immediate to see that the rank of the PBH matrix is n in $\mathbb{C} \setminus \{0\}$ if and only if all the eigenvalues of the matrix $\bar{F}_{22}$ are zero. ∎

6.9 Bibliography

For the reader interested about canonical and invariant forms we suggest

1. S. MacLane, G.Birkhoff "*Algebra*" MacMillan, 1967, pages from 287 on

To deepen a large part of the arguments treated in this chapter one may also check

2. T.Kailath "*Linear Systems*" Prentice-Hall, 1980

An alternative proof of Heymann's lemma is given in

3. M.L.J.Hautus "*A simple proof of Heymann's lemma*" IEEE Transactions on Automatic Control, 1977, pp. 885-6

As for the basis transformation inducing multivariable controllable canonical forms and for Rosenbrock's theorem see

4. D.Luenberger "*Canonical forms for linear multivariable systems*" IEEE Trans AC, 1967, pp.290-3

5. H.H. Rosenbrock "*State space and multivariable theory*" Wiley, 1970